Methods in Human Geography

Quantitative Methods: Spatial Analysis II



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This week

Part I

- Mapping 101.

Part II

- More spatial analysis: a tour of what else is out there.
- Module summary.

Mapping 101

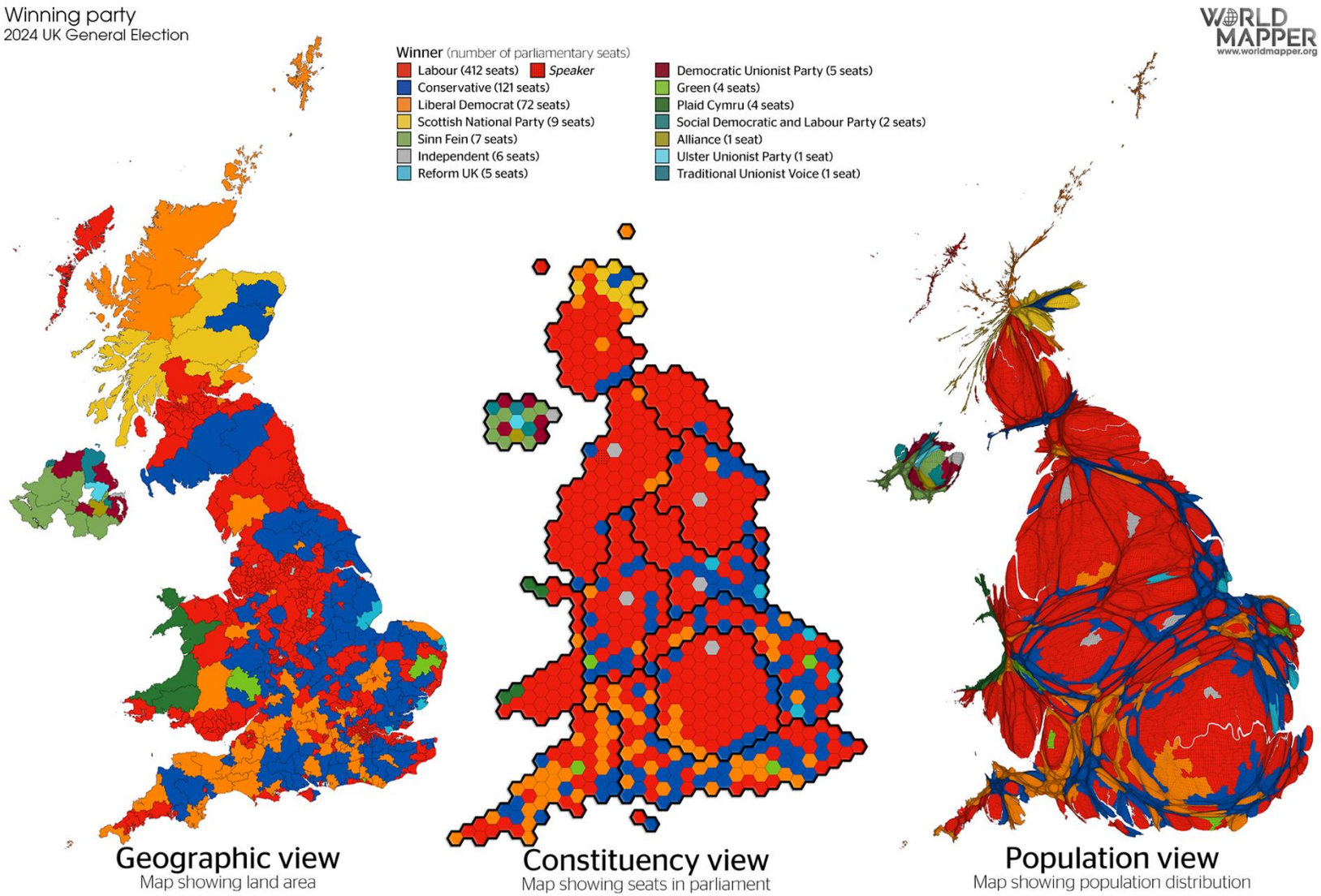
Mapping 101

- Use tables for precision.
- Use graphs to effectively summarise large volumes of information.
- Use maps to show spatially heterogeneous information.

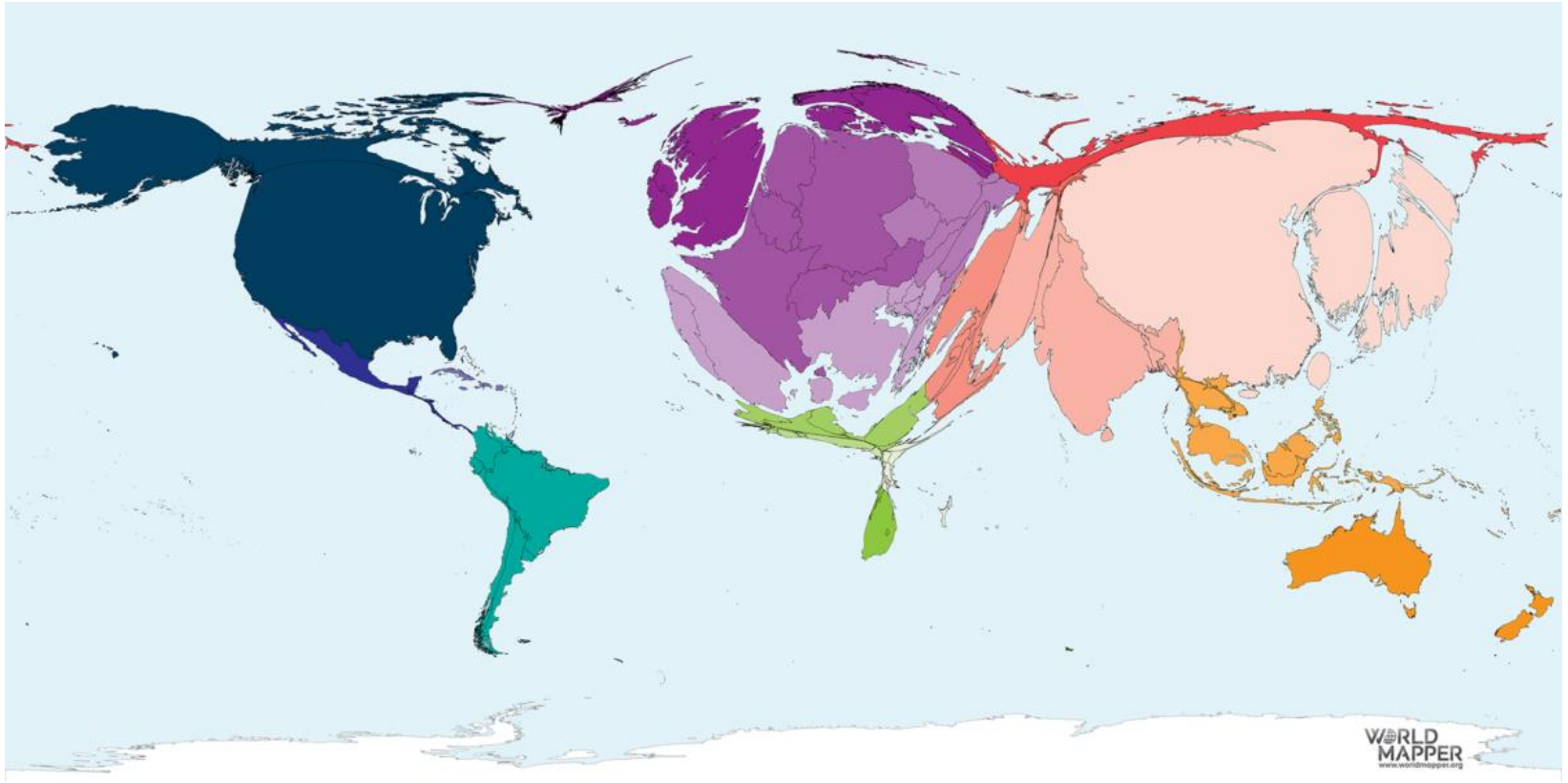
Mapping 101

- Mapping is a process and a way of thinking which can conceal and reveal.
- It requires simplification and choices, so a map is never an ‘objective’ representation.
- Most common: choropleth map, which shades each spatial unit by its aggregate value.
- It is hard to make a good map, but relatively easy to make a bad one.

Mapping 101

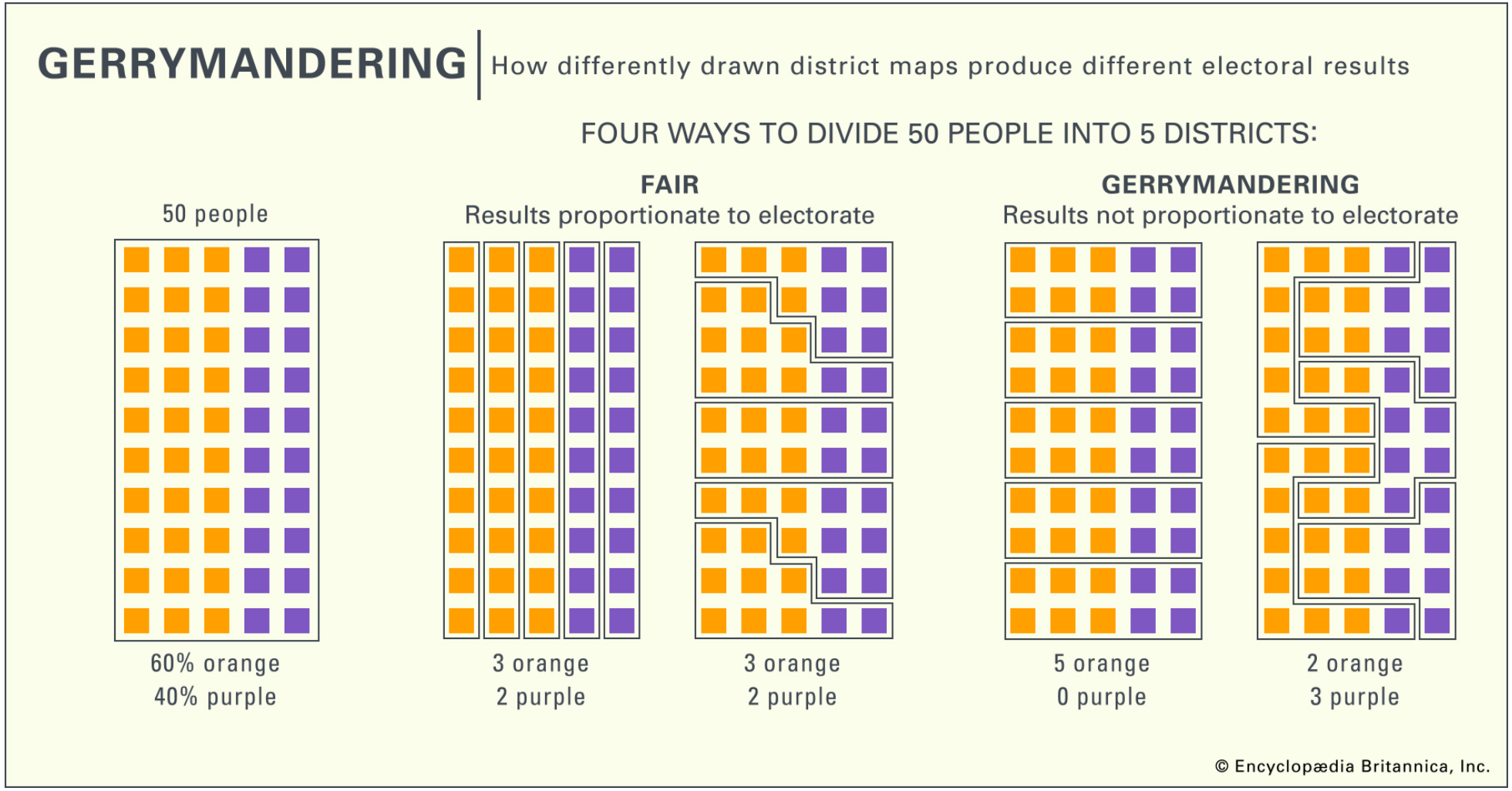

Winning party
2024 UK General Election
WORLD MAPPER
www.worldmapper.org
Winner (number of parliamentary seats)
Labour (412 seats)
Speaker
Conservative (121 seats)
Liberal Democrat (72 seats)
Scottish National Party (9 seats)
Sinn Fein (7 seats)
Independent (6 seats)
Reform UK (5 seats)
Democratic Unionist Party (5 seats)
Green (4 seats)
Plaid Cymru (4 seats)
Social Democratic and Labour Party (2 seats)
Alliance (1 seat)
Ulster Unionist Party (1 seat)
Traditional Unionist Voice (1 seat)
Geographic view
Map showing land area
Constituency view
Map showing seats in parliament
Population view
Map showing population distribution

Mapping 101



Scientific papers published in 2016.

Mapping 101



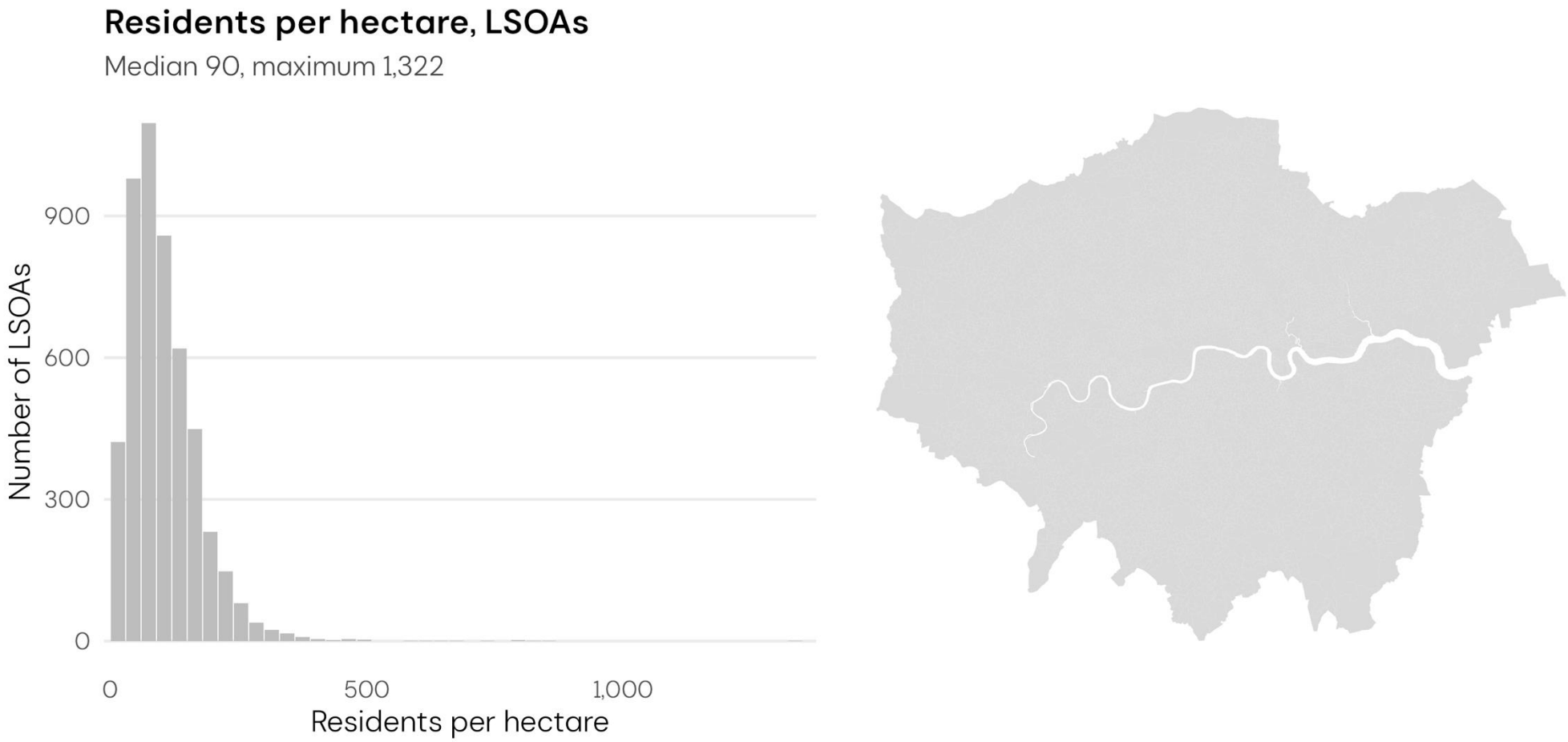
Encyclopaedia Britannica, Inc. Gerrymandering. [Online]
https://www.britannica.com/topic/gerrymandering

Mapping 101

Typical workflow:

- Importing spatial data into your GIS.
- Importing attribute data into your GIS – cleaning and wrangling might be required prior to importing (e.g. data preparation in R).
- Geoprocessing (e.g. point in polygon, but many more methods!).
- Creating and modifying maps and cartographic outputs.

Mapping 101

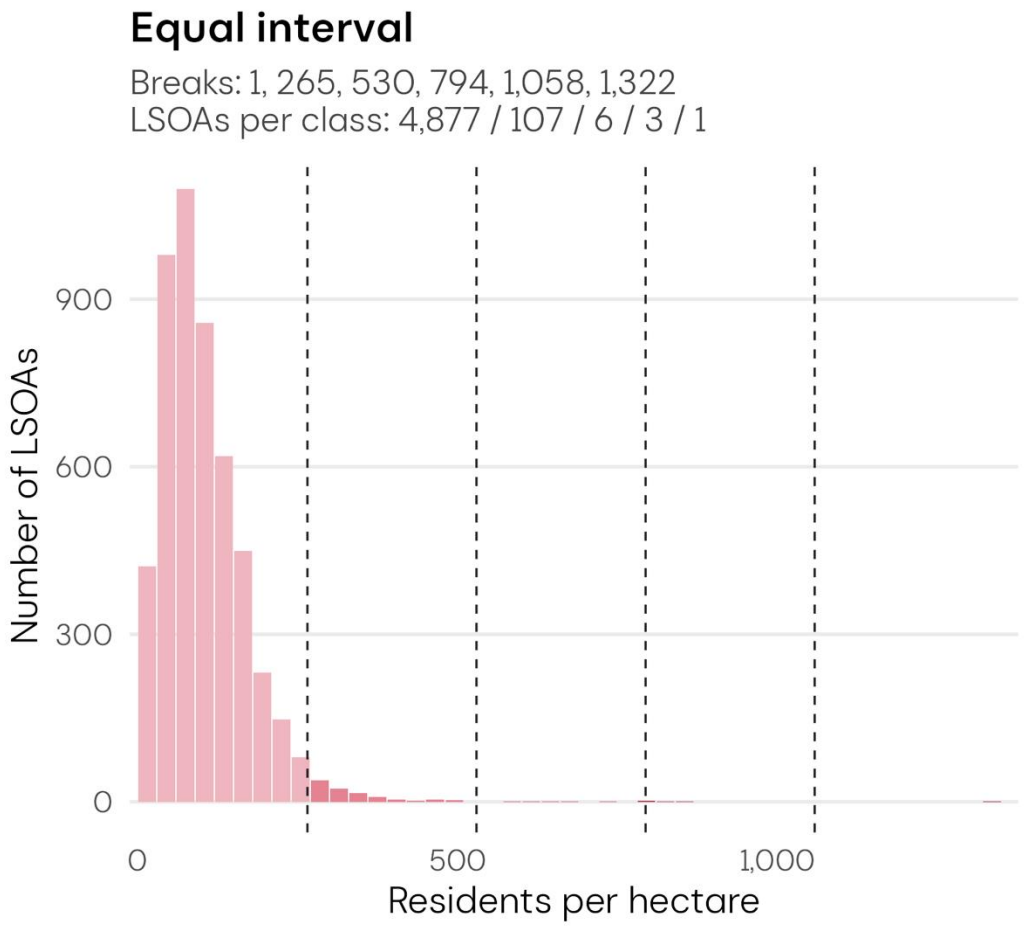


Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Mapping 101

Equal Interval: This method divides the range of data values into equal intervals based on numerical value. If the data distribution is skewed, this approach can give more weight to outliers, potentially leading to less balanced representation across the classes.

Mapping 101



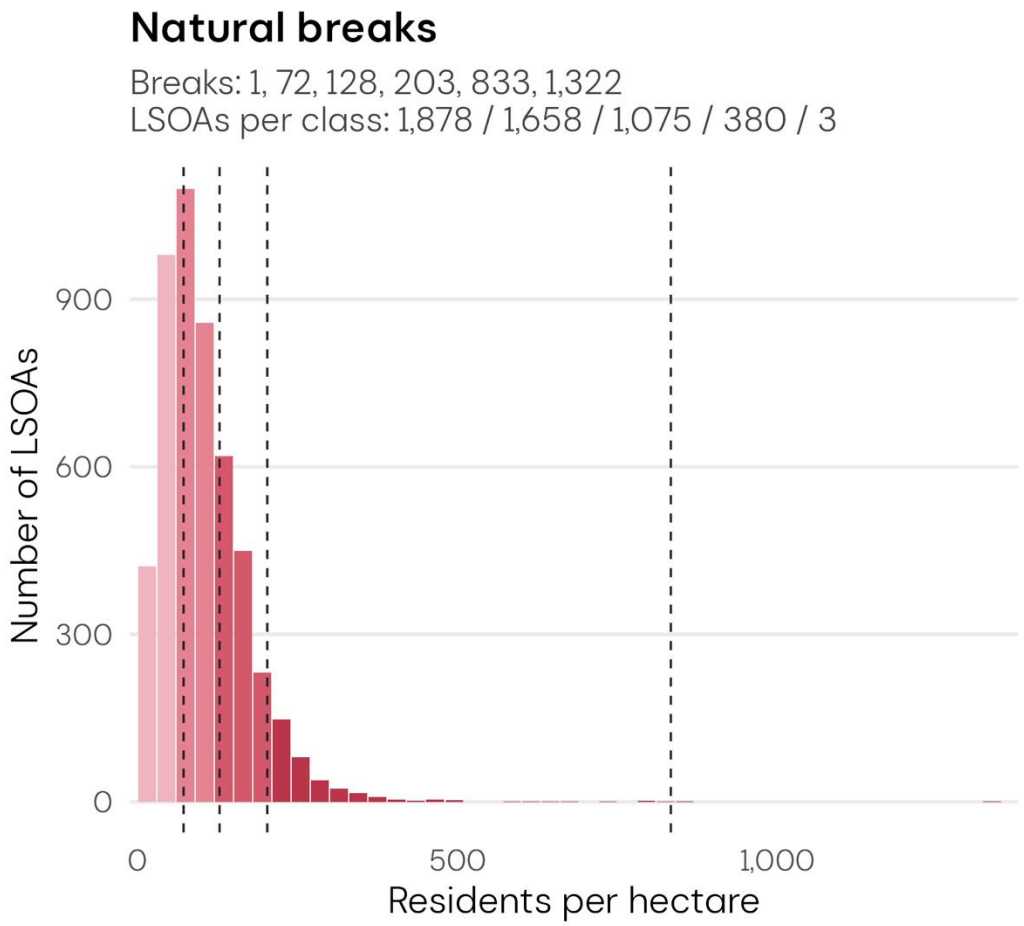


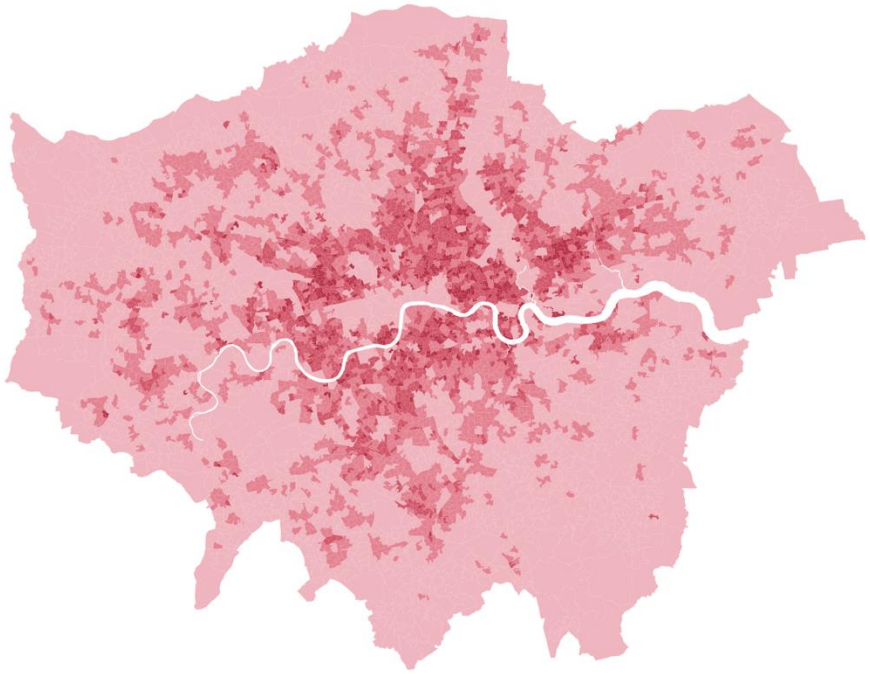
Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Mapping 101

Natural Breaks/Jenks: This method aims to minimise the average deviation within each class from the class mean while maximising the deviation between the means of different classes. In the case of a skewed distribution, outliers often end up in their own class, although this can vary depending on the number of classes chosen.

Mapping 101



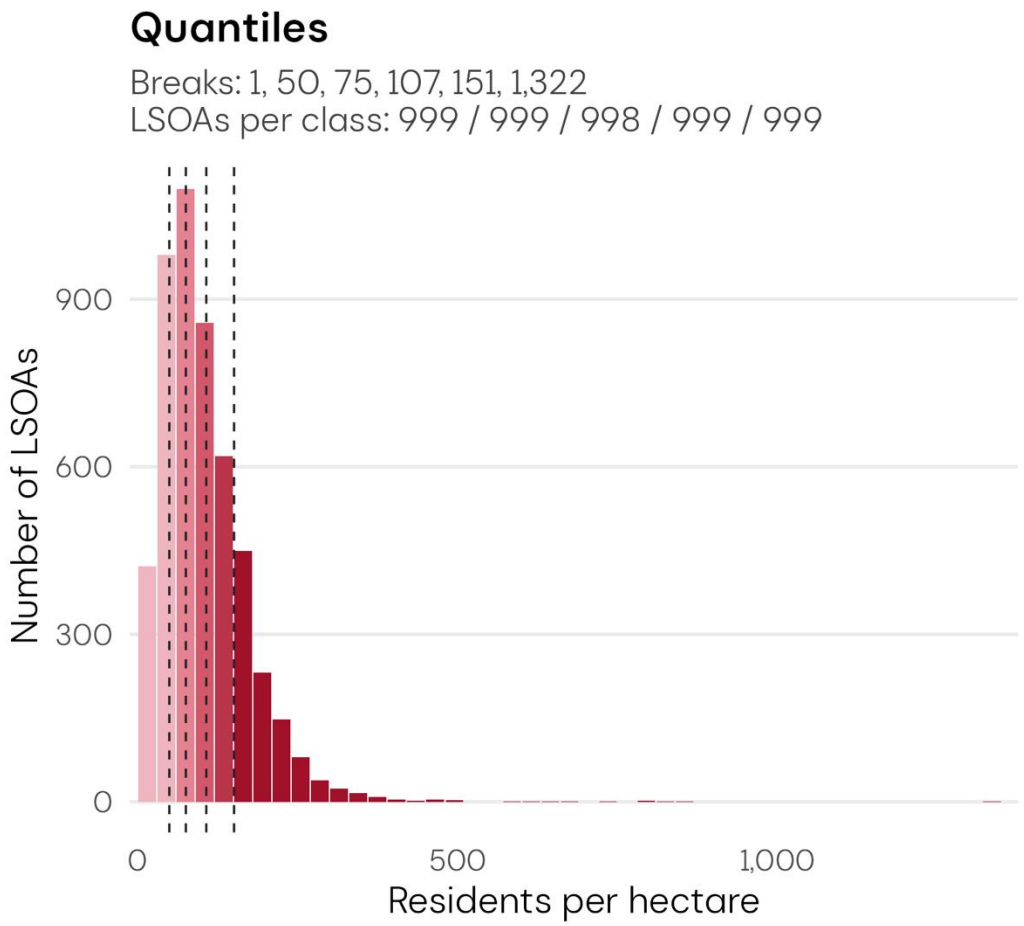


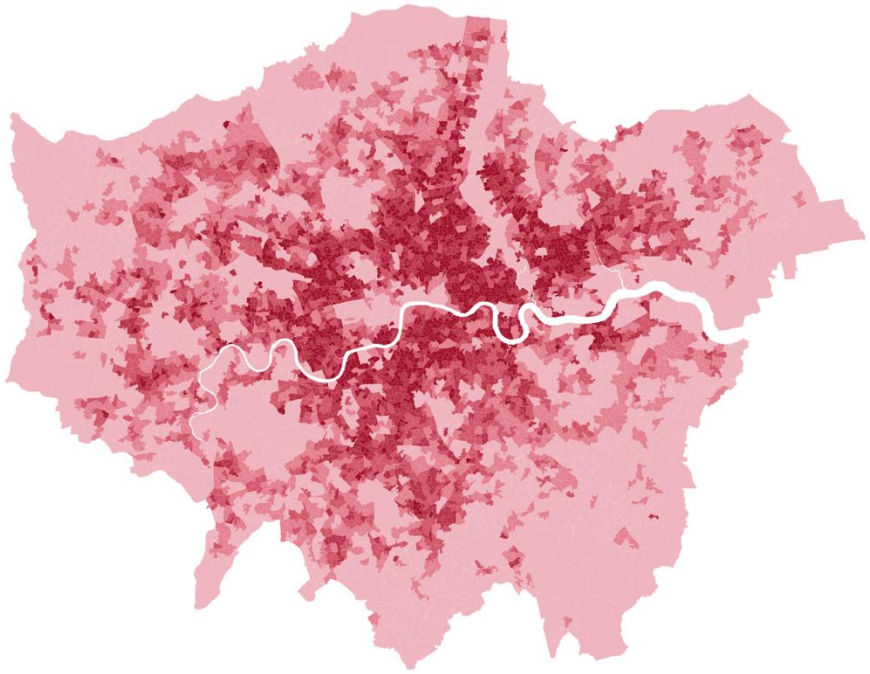
Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Mapping 101

Quantiles: This method divides the distribution into classes so that each bin contains the same number of values. The splits are based on the rank order of the data rather than the actual values. When the distribution is skewed, this approach can result in very different values being grouped together in the same bin.

Mapping 101



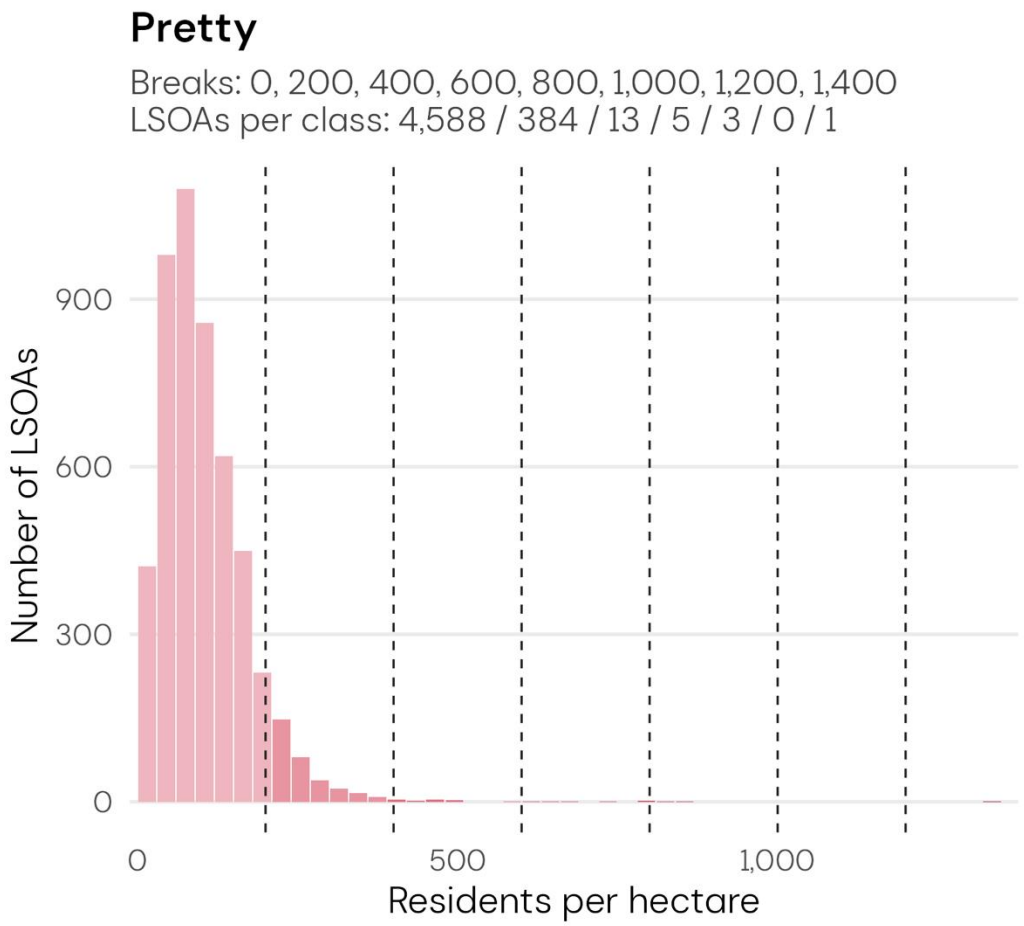


Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Mapping 101

Pretty: This method divides the distribution using a series of aesthetically pleasing, rounded values. The splits are not influenced by the data or its distribution. In cases of skewed distribution, this approach can lead to bins containing a low number of values, potentially overlooking important data patterns.

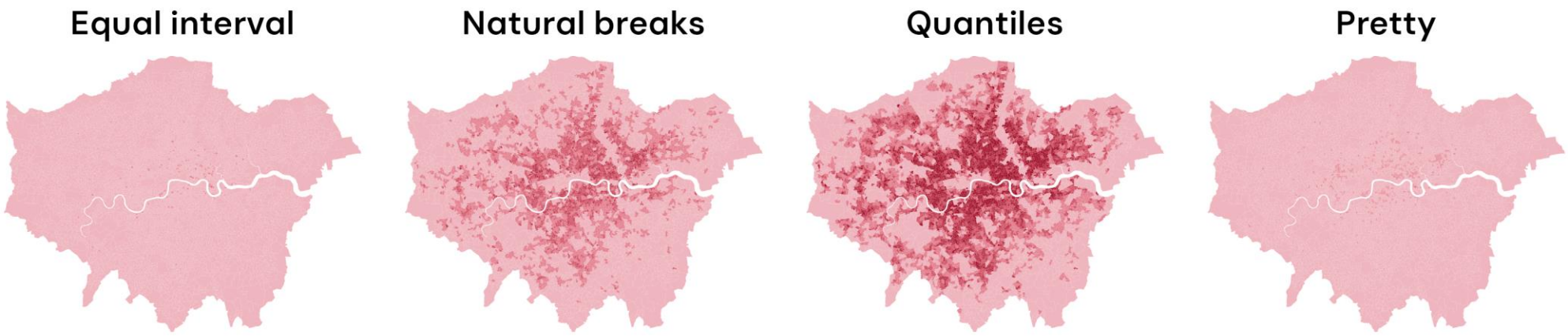
Mapping 101





Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Different classifications, different maps

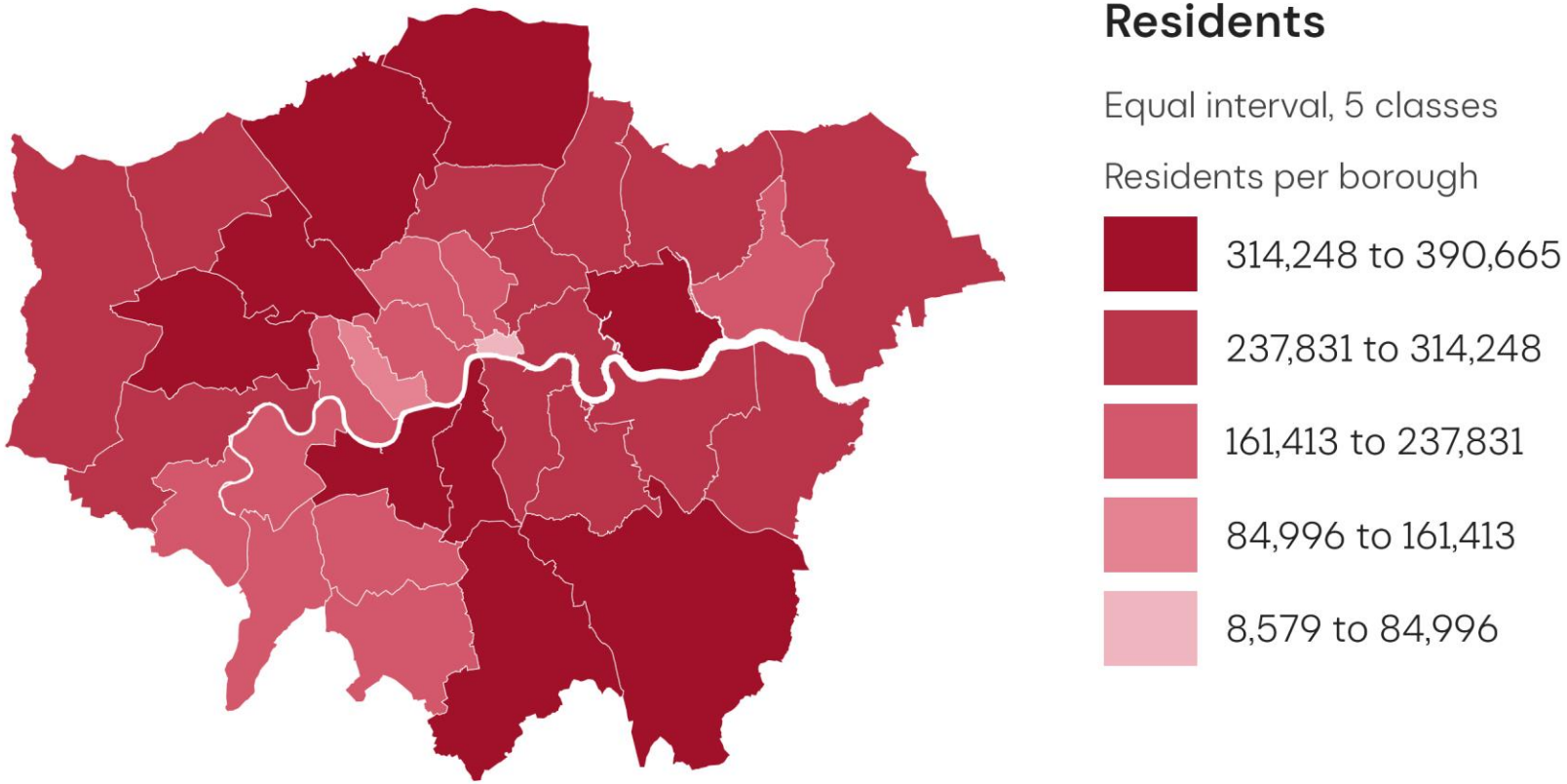


Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Mapping 101

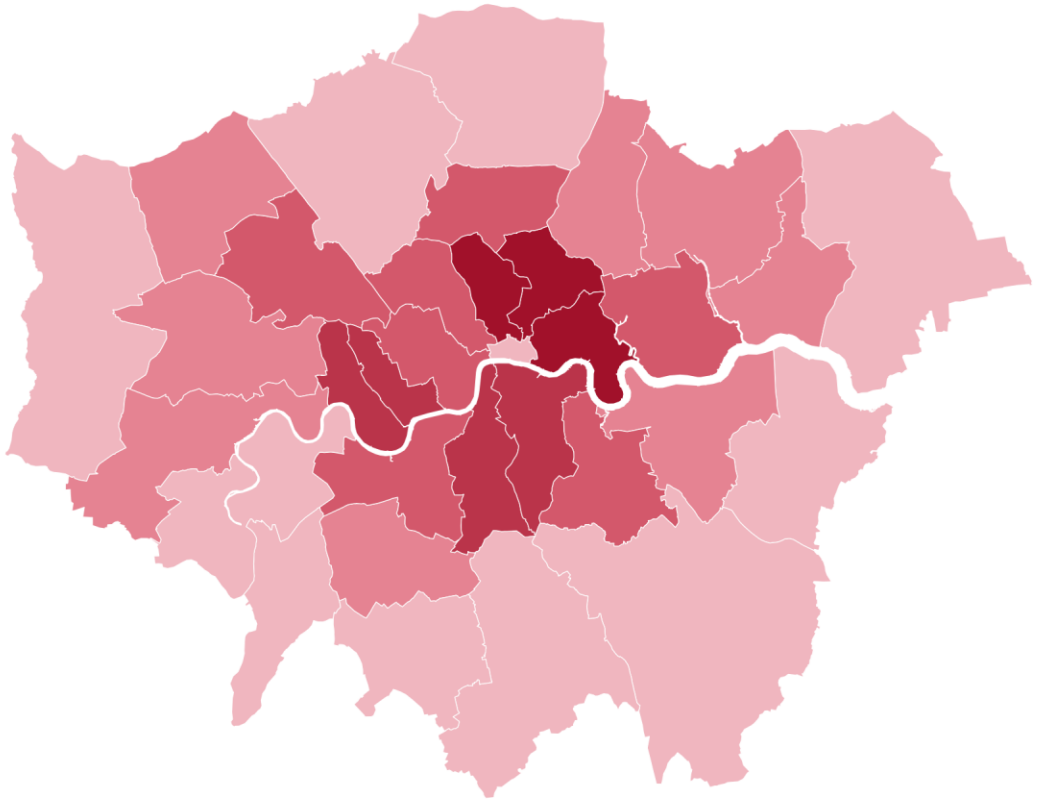
Standardisation: Plotting raw numbers on choropleth maps should generally be avoided, as it can be misleading. Standardisation is achieved by dividing the variable of interest (numerator) by a standardising variable (denominator), such as total population, area, or another relevant metric. This approach allows for more accurate and meaningful comparisons across different regions or units.

Mapping 101



Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Mapping 101



Residents per hectare

Equal interval, 5 classes

Residents per hectare

- 130 to 157
- 103 to 130
- 76 to 103
- 49 to 76
- 22 to 49

Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

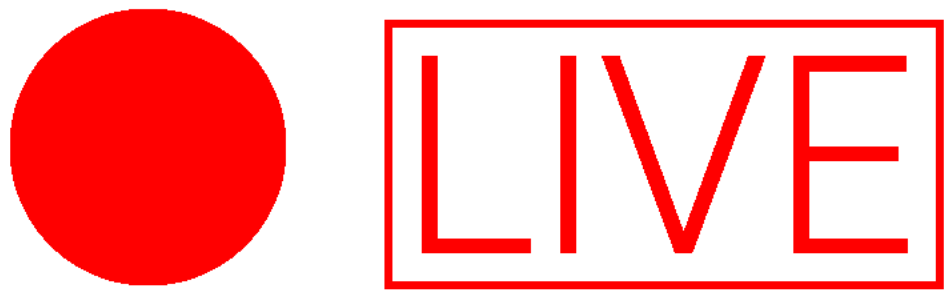
Mapping 101

Publishable maps should include the following essential elements:

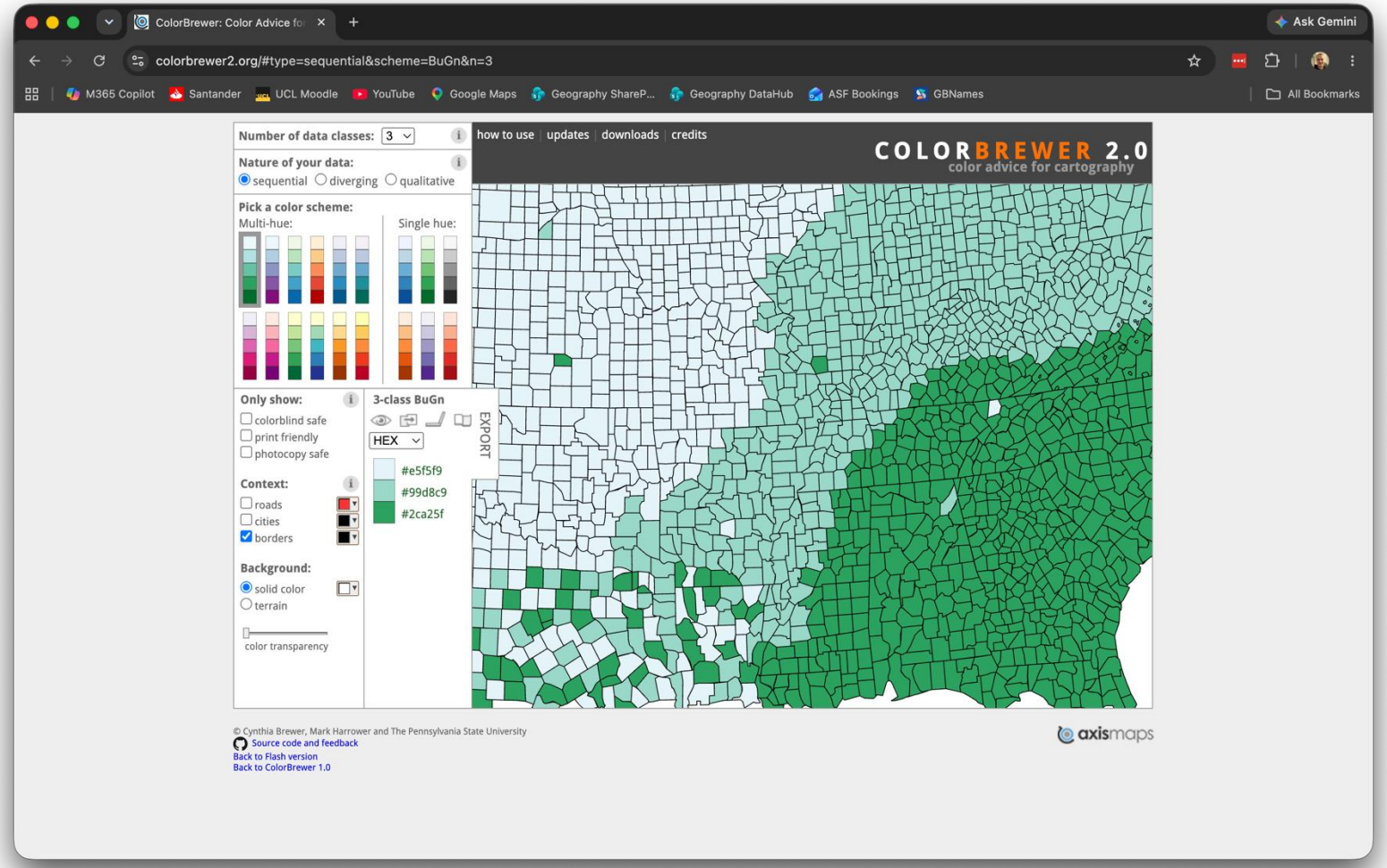
- Title
- Legend
- Scale bar / scale text
- North arrow
- And: attribution and data sources

Optional:

- Text and labels can be important for providing context, but may sometimes be distracting, depending on the map's purpose.


LIVE

Mapping 101

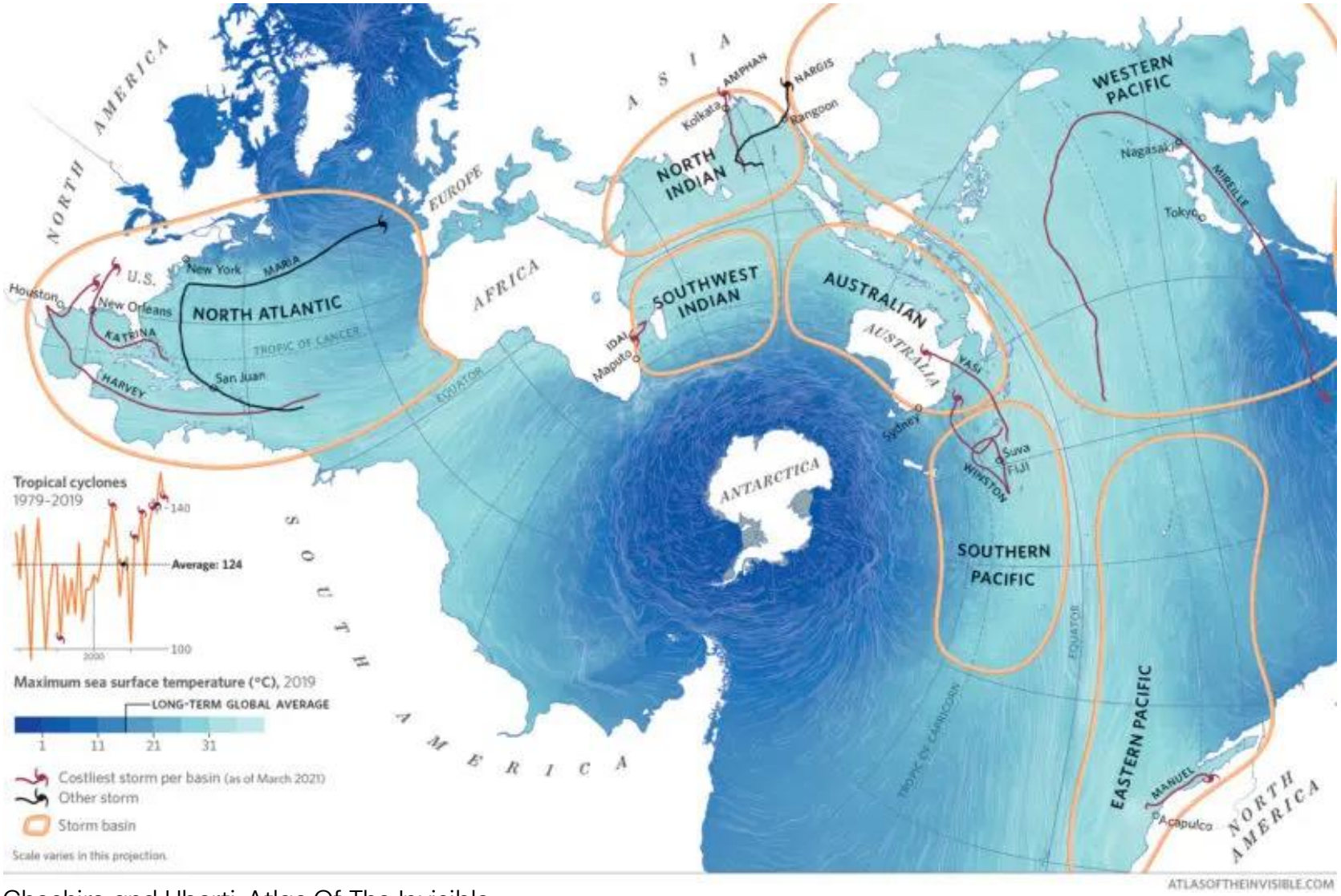

ColorBrewer: Color Advice fo
Ask Gemini
colorbrewer2.org/#type=sequential&scheme=BuGn&n=3
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
Number of data classes: 3
Nature of your data:
sequential
diverging
qualitative
Pick a color scheme:
Multi-hue:
Single hue:
Only show:
colorblind safe
print friendly
photocopy safe
Context:
roads
cities
borders
Background:
solid color
terrain
color transparency
3-class BuGn
EXPORT
HEX
#e5f5f9
#99d8c9
#2ca25f
how to use | updates | downloads | credits
COLORBREWER 2.0
color advice for cartography
© Cynthia Brewer, Mark Harrower and The Pennsylvania State University
Source code and feedback
Back to Flash version
Back to ColorBrewer 1.0
axismaps

Cartography



Cheshire and Uberti. Atlas Of The Invisible

Cartography



Cheshire and Uberti. Atlas Of The Invisible

Mentimeter

- Go to www.menti.com.
- Use code: XXXX XXXX

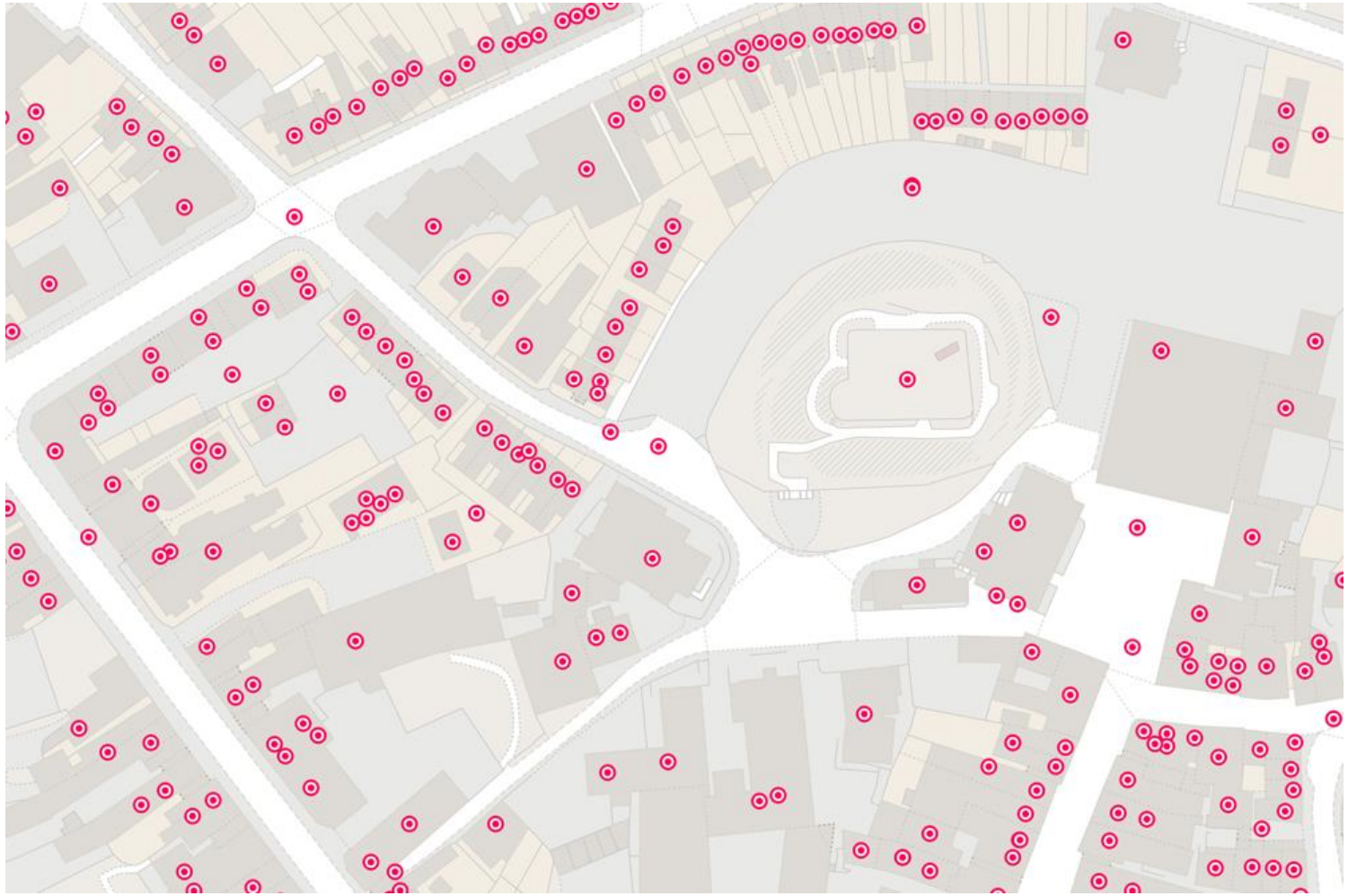
Break

More spatial analysis

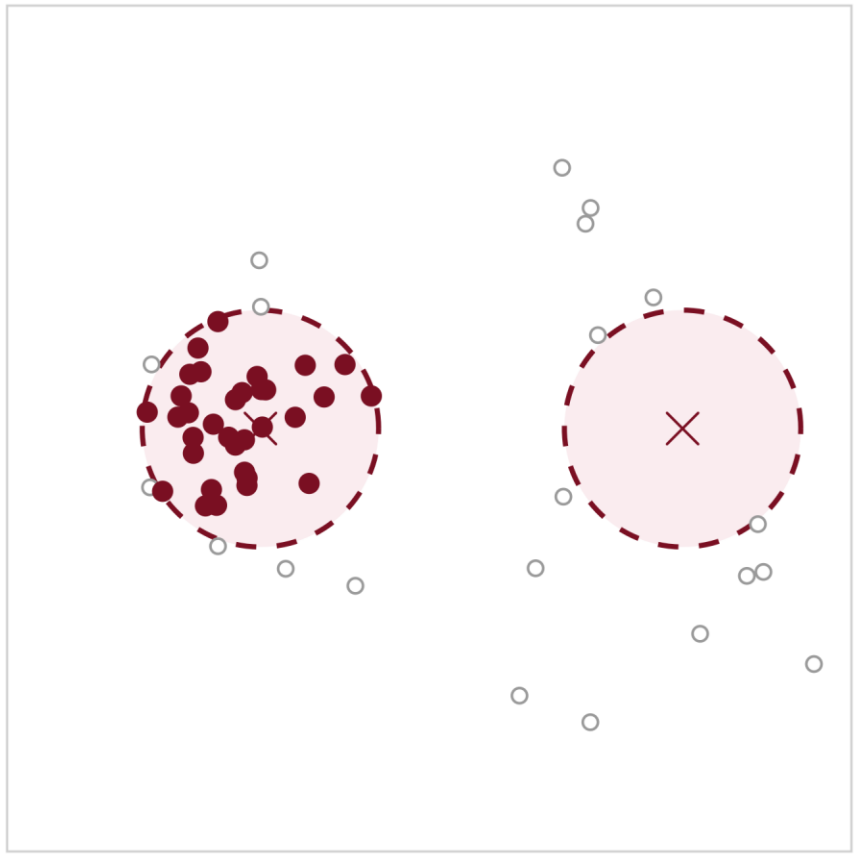
Geocoding

Forename	Surname	Address
Justin	van Dijk	Flat 18 Terry House SW22NT London

Geocoding



Fixed kernel: the same distance everywhere



× regression point

● point inside the window

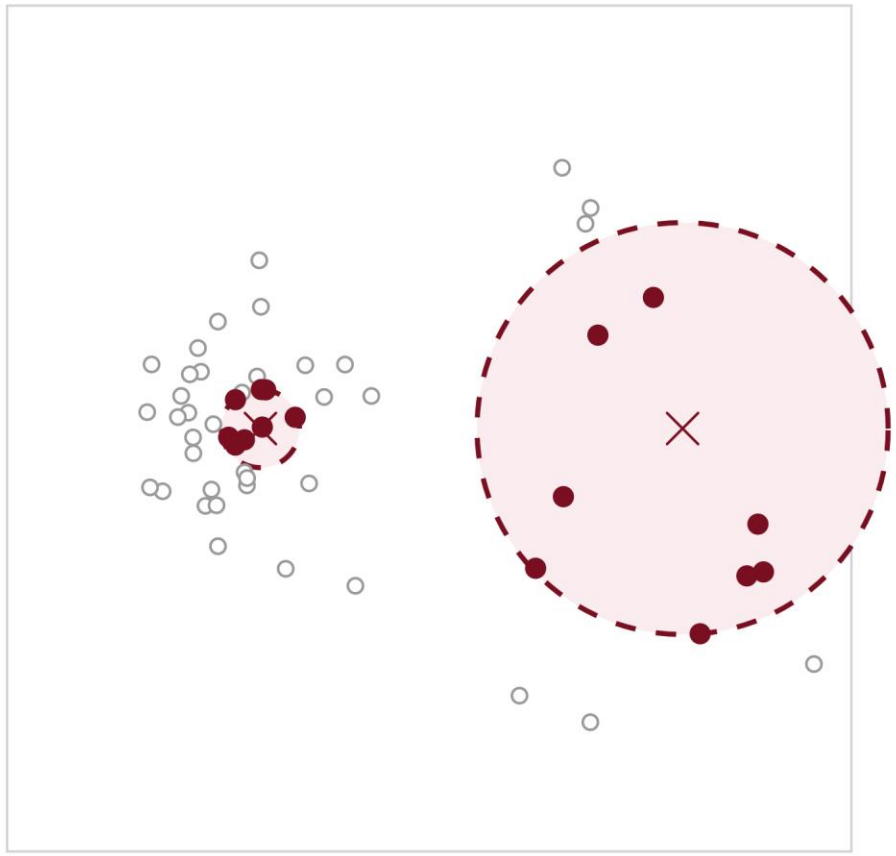
○ point outside

Same radius everywhere.

Dense area: 33 points in the window.

Sparse area: 0 points in the window.

Adaptive kernel: the same number of neighbours



× regression point

● point inside the window

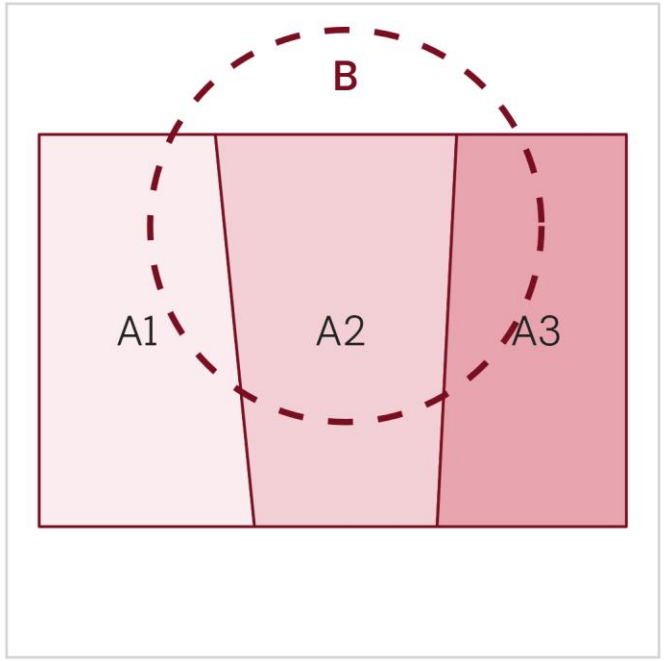
○ point outside

Same number of points: 8.

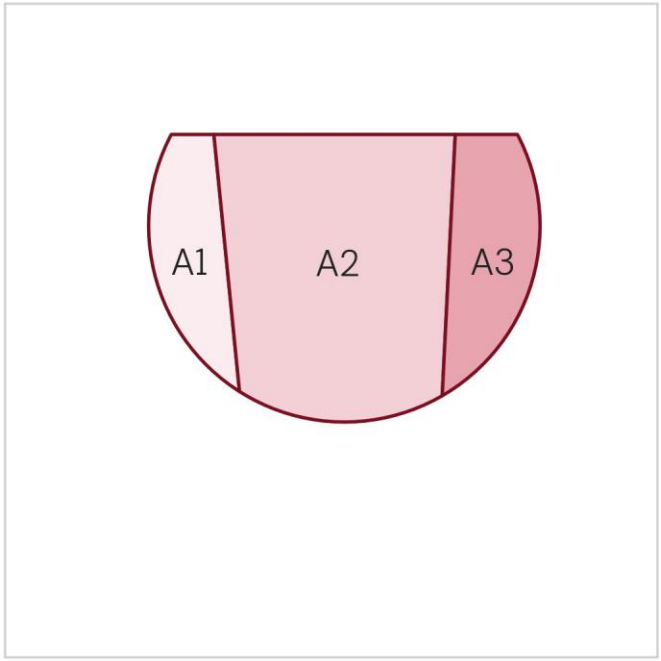
Dense area: a small window.

Sparse area: a window 5.3 times wider.

Clip: cut one layer to the shape of another



Layer A and layer B

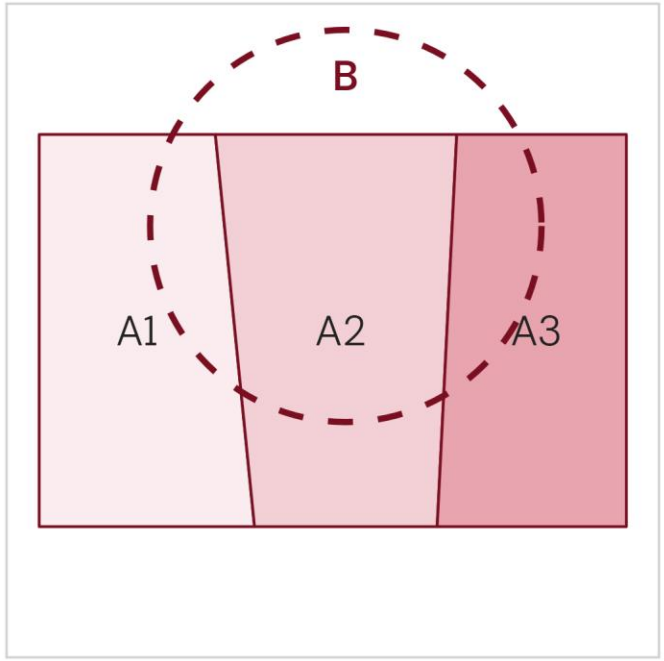


Result

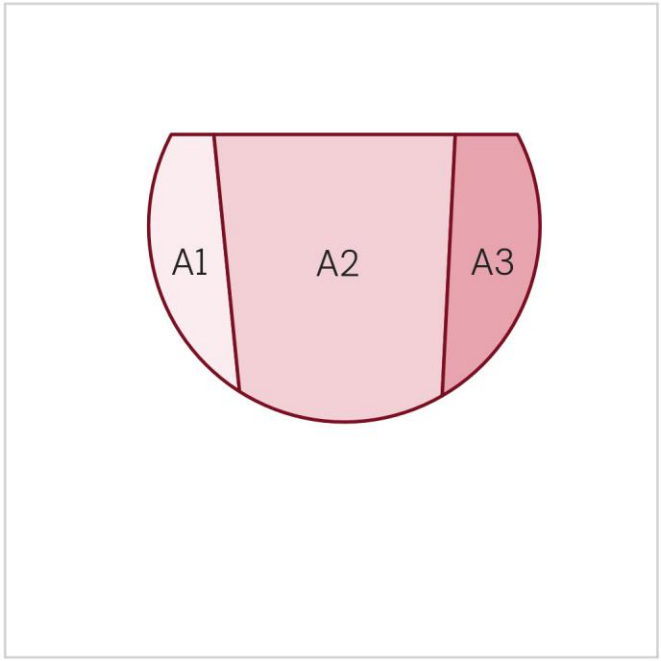
name
A1
A2
A3

Attribute table

Intersect: keep only where two layers overlap



Layer A and layer B

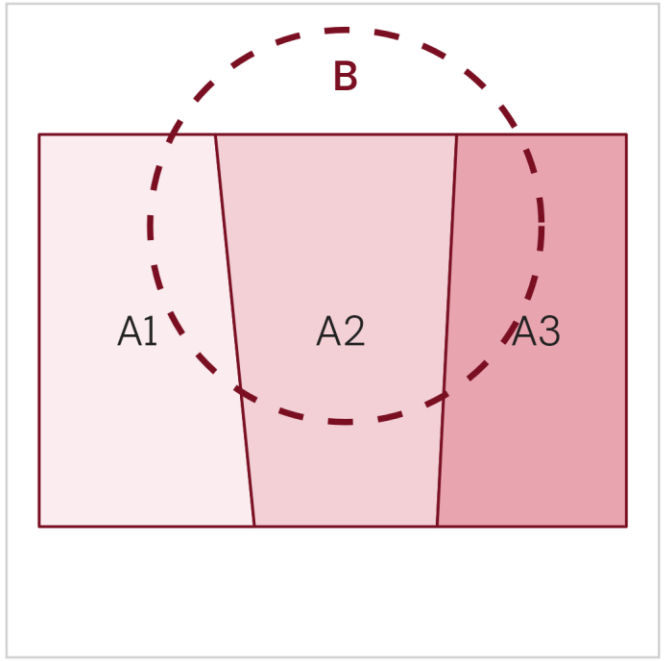


Result

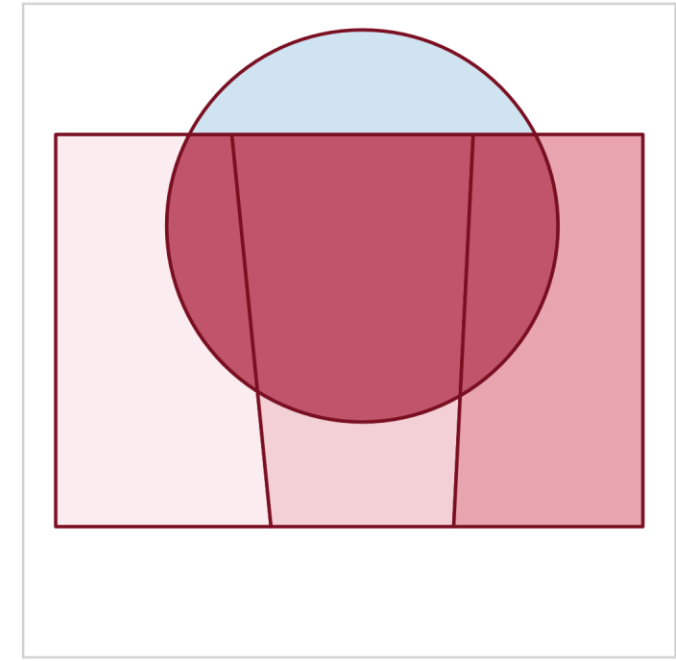
name	zone
A1	B
A2	B
A3	B

Attribute table

Union: combine two layers, keep every area



Layer A and layer B

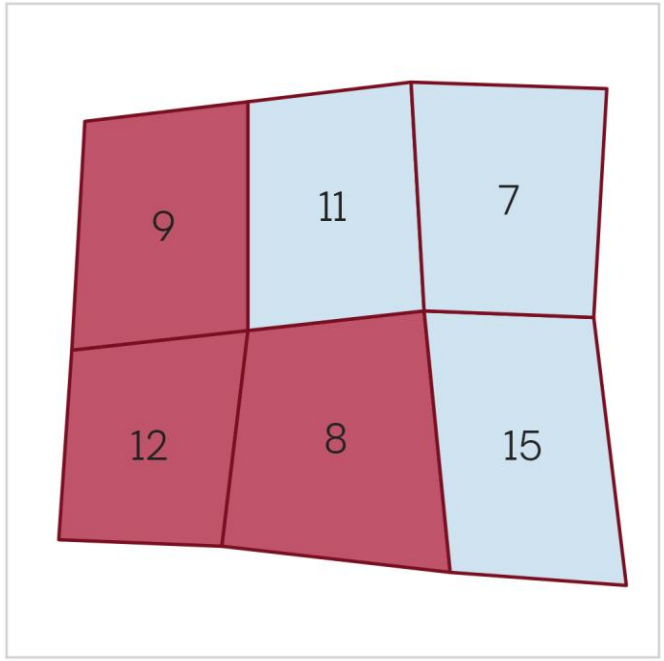


Result

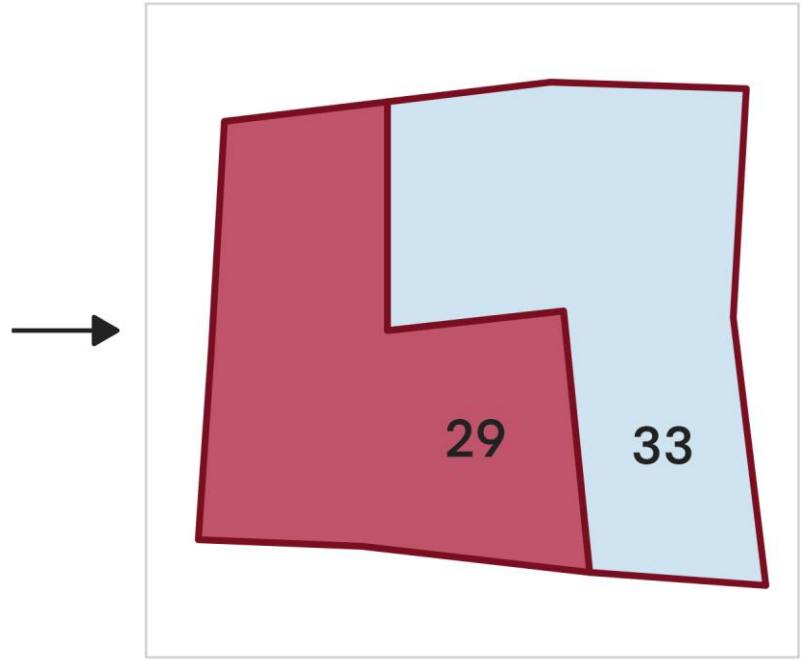
name	zone
A1	B
A2	B
A3	B
A1	NA
A2	NA
A3	NA
NA	B

Attribute table

Dissolve: merge features into one



Six features with a count

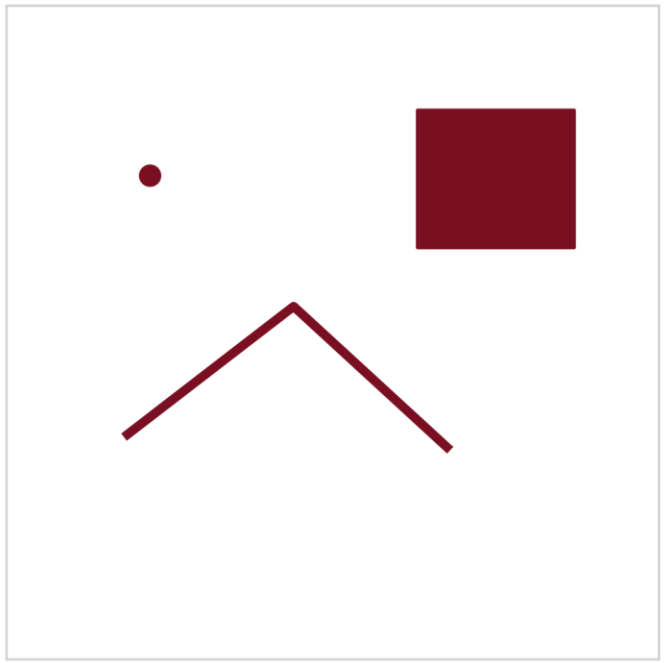


Dissolved by region, counts summed

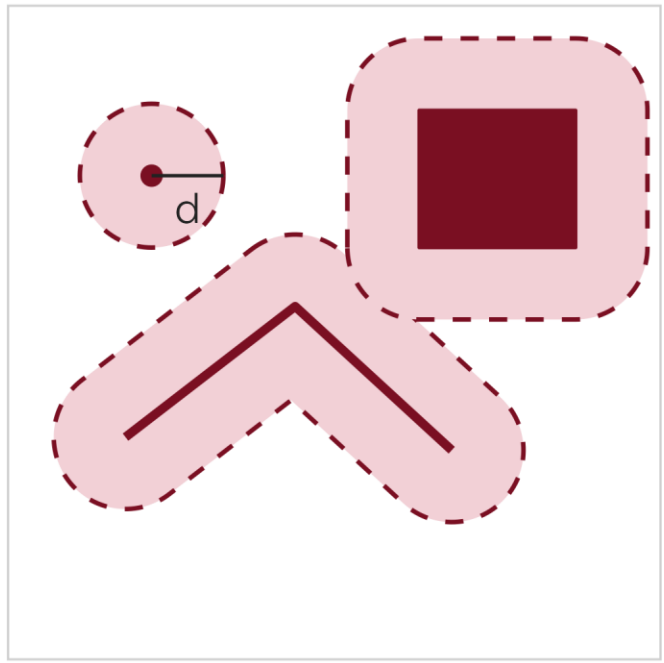
region	count
X	29
Y	33

Attribute table

Buffer: draw a zone of fixed distance



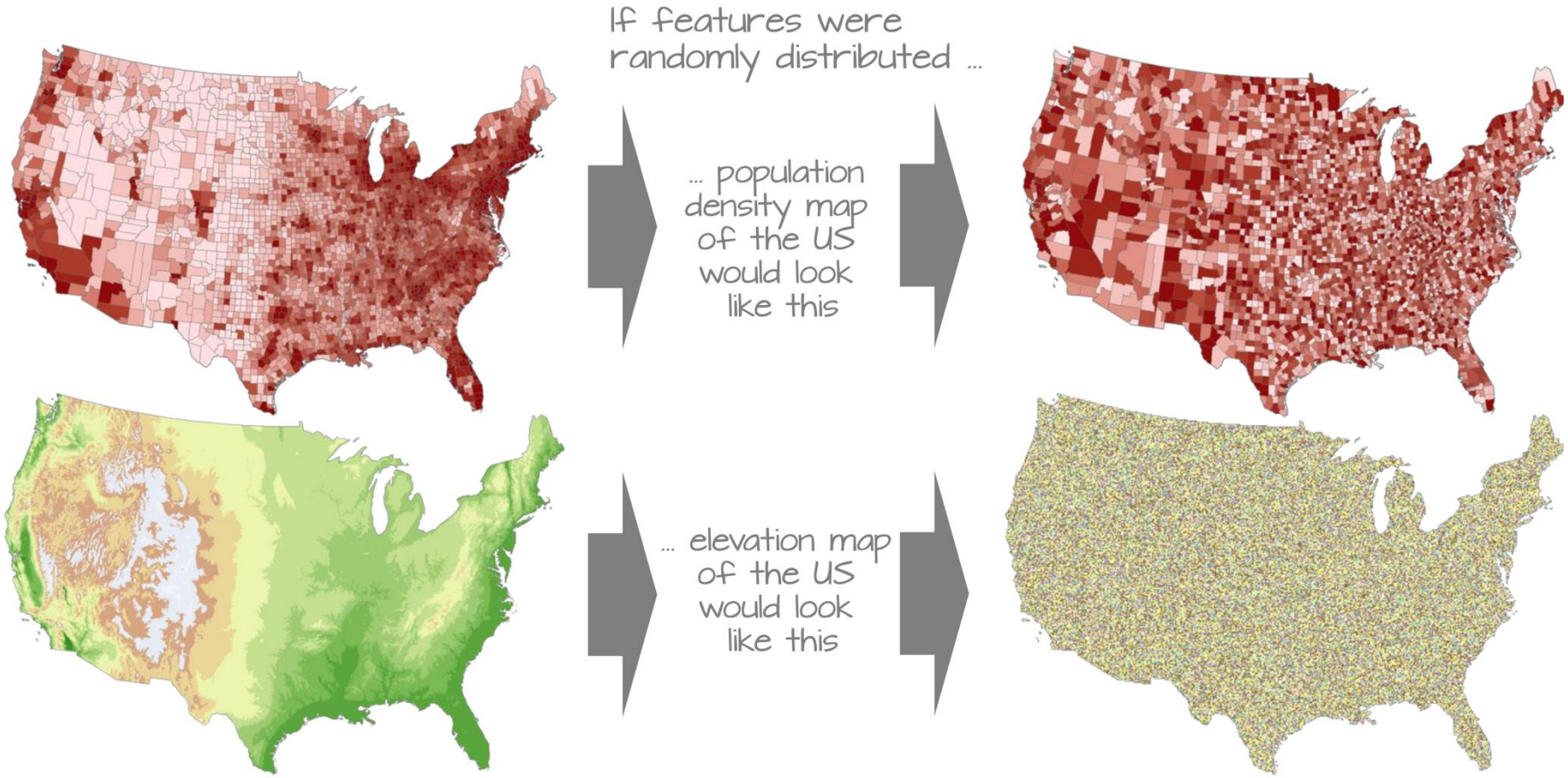
A point, a line and a polygon



Buffers of distance d

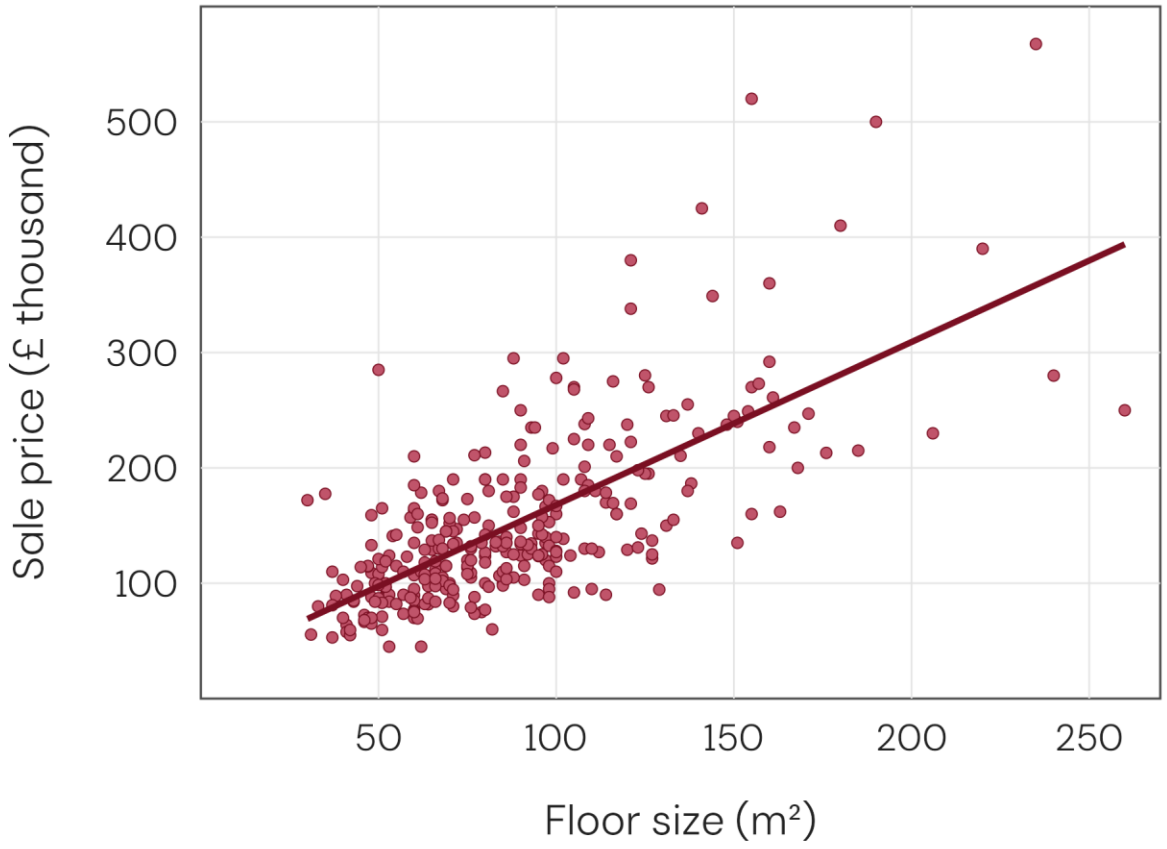
Every buffer reaches the same distance d from its feature, whatever the shape.

Spatial dependence



Gimond, M. 2021. Intro to GIS and Spatial Analysis. [online]
https://mgimond.github.io/Spatial/introGIS.html

One slope for the whole of London



One line for all of London

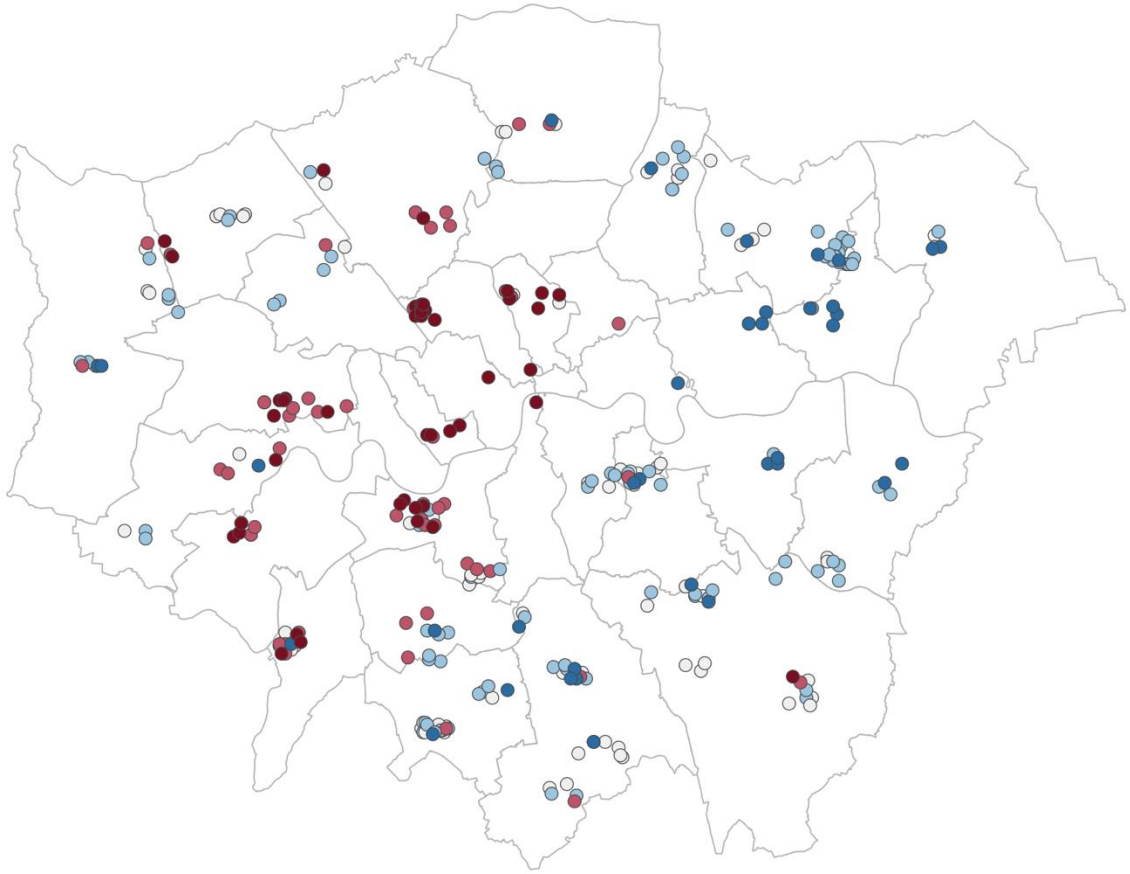
Each extra square metre goes with about £1,412 more.

In the centre and in the outer boroughs alike.

$R^2 = 0.49$

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

Residuals can cluster in space



OLS residual (£ thousand)

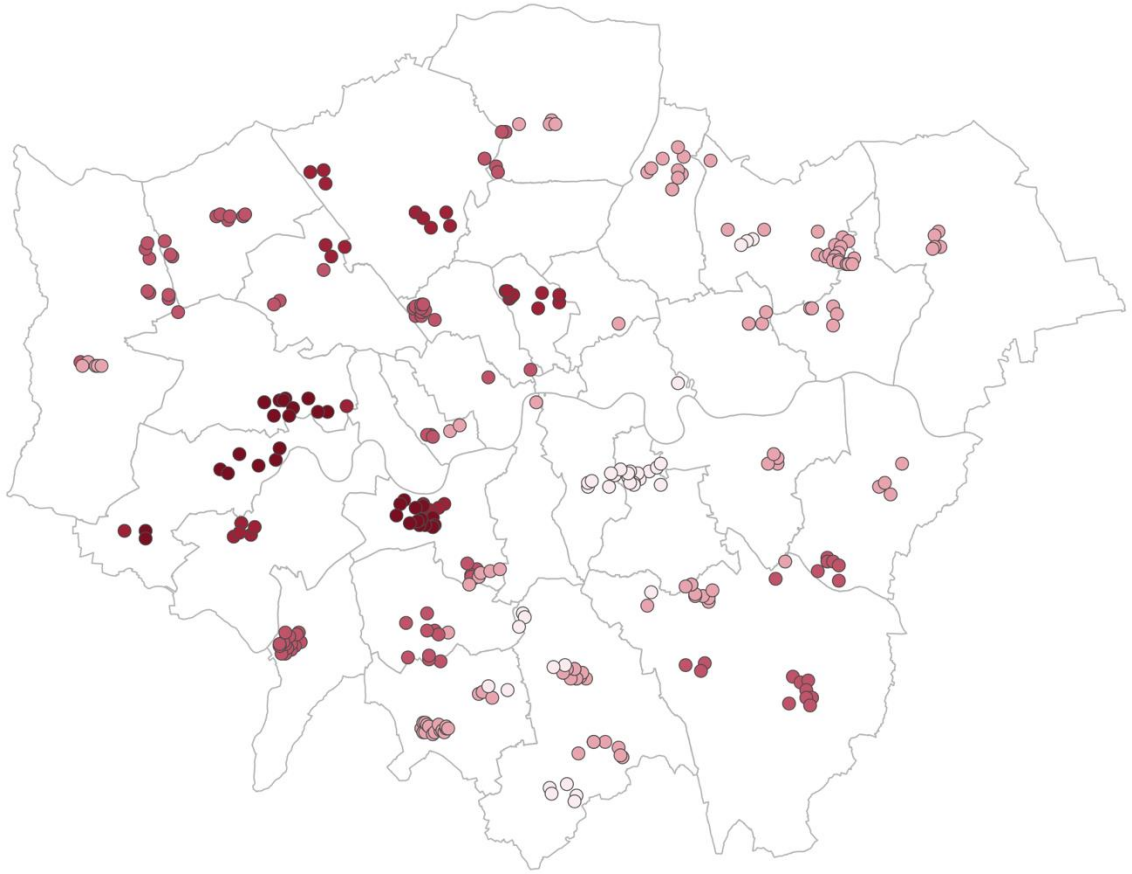
- below −50
- −50 to −15
- −15 to 15
- 15 to 50
- above 50

Red: sold for more than the line predicts.

Blue: sold for less.

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

Local slopes: the price of space varies by place



Local slope (£ per m²)

- below 1,000
- 1,000 to 1,400
- 1,400 to 2,000
- 2,000 to 2,600
- above 2,600

Local slopes run from £762 to £3,079.

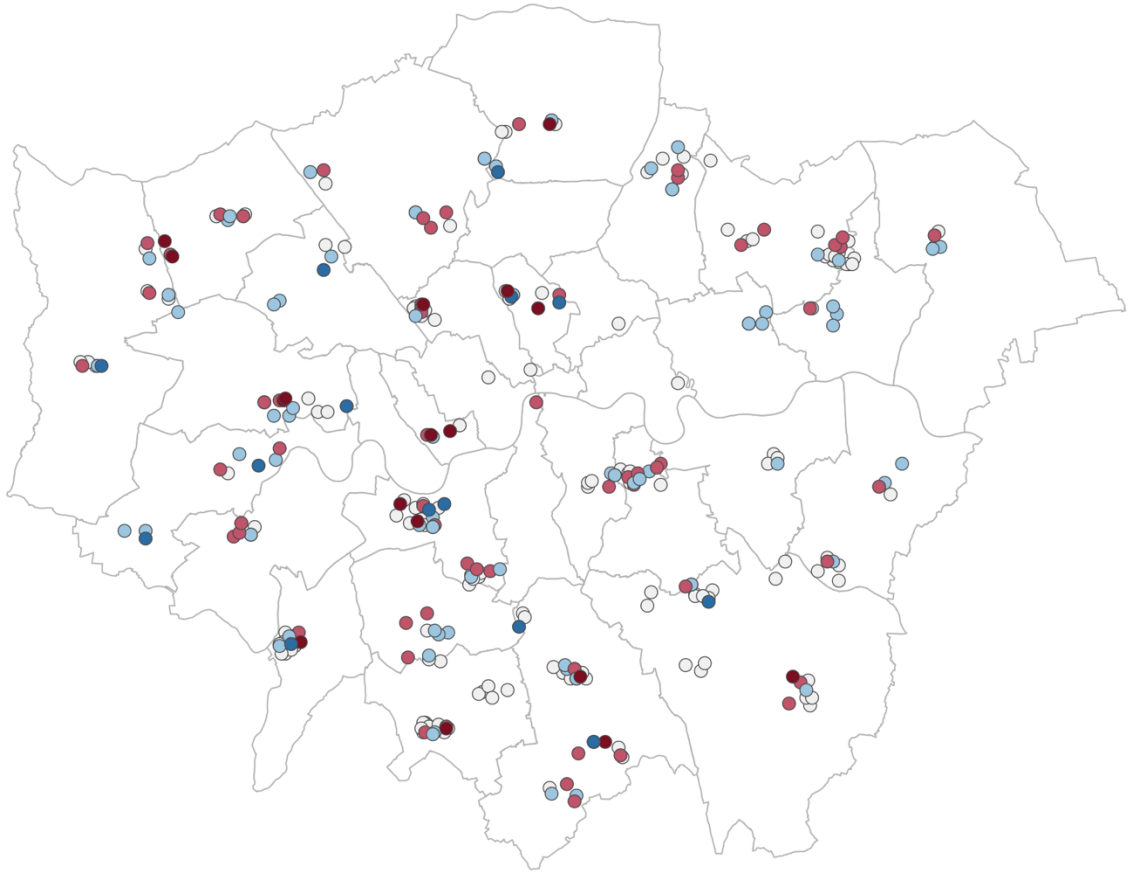
The global slope was £1,412.

Local R^2:

0.54 to 0.95

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

After GWR: the residuals no longer cluster



GWR residual (£ thousand)

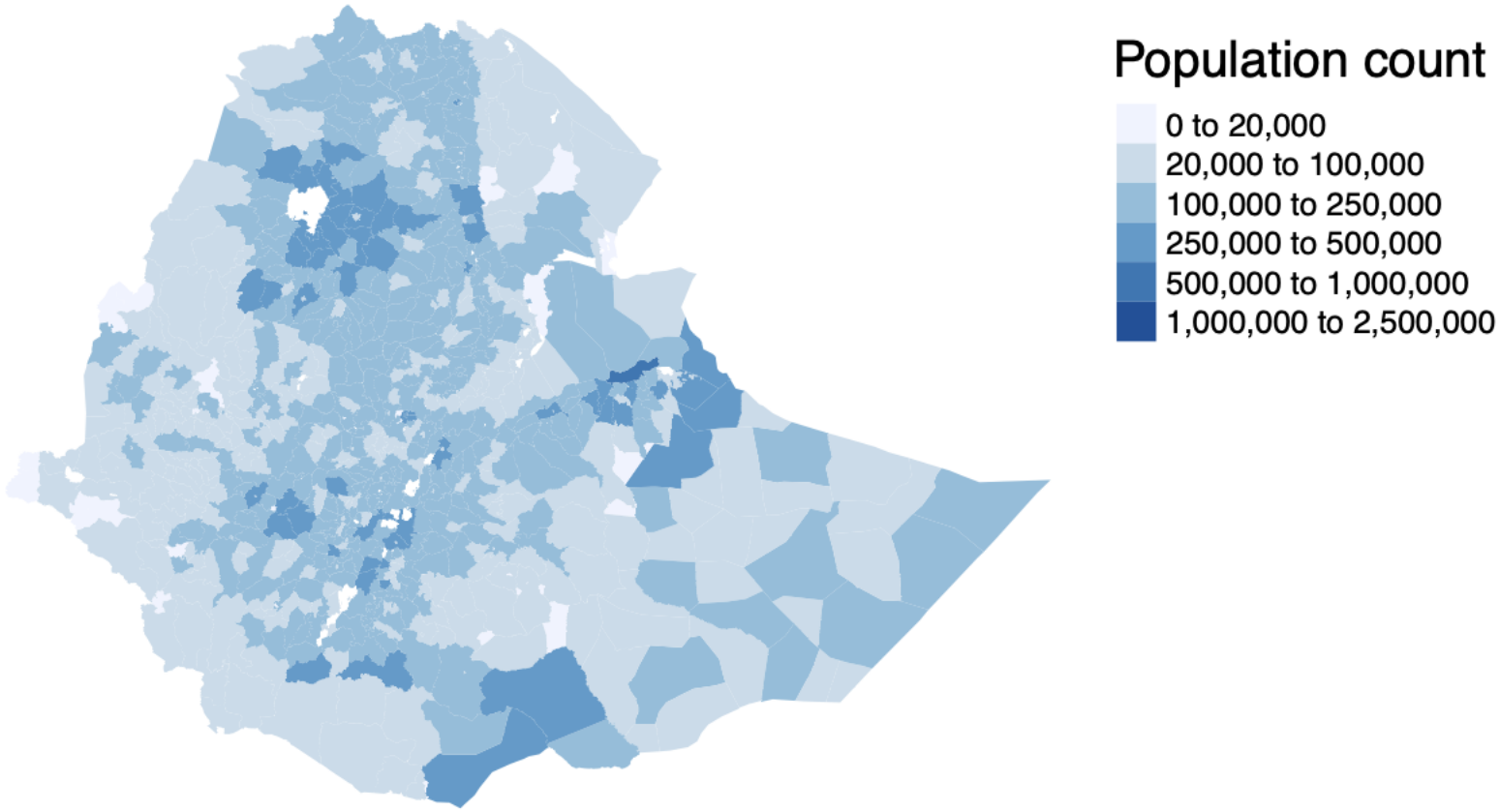
- below −50
- −50 to −15
- −15 to 15
- 15 to 50
- above 50

R^2 rises from 0.49 to 0.84.

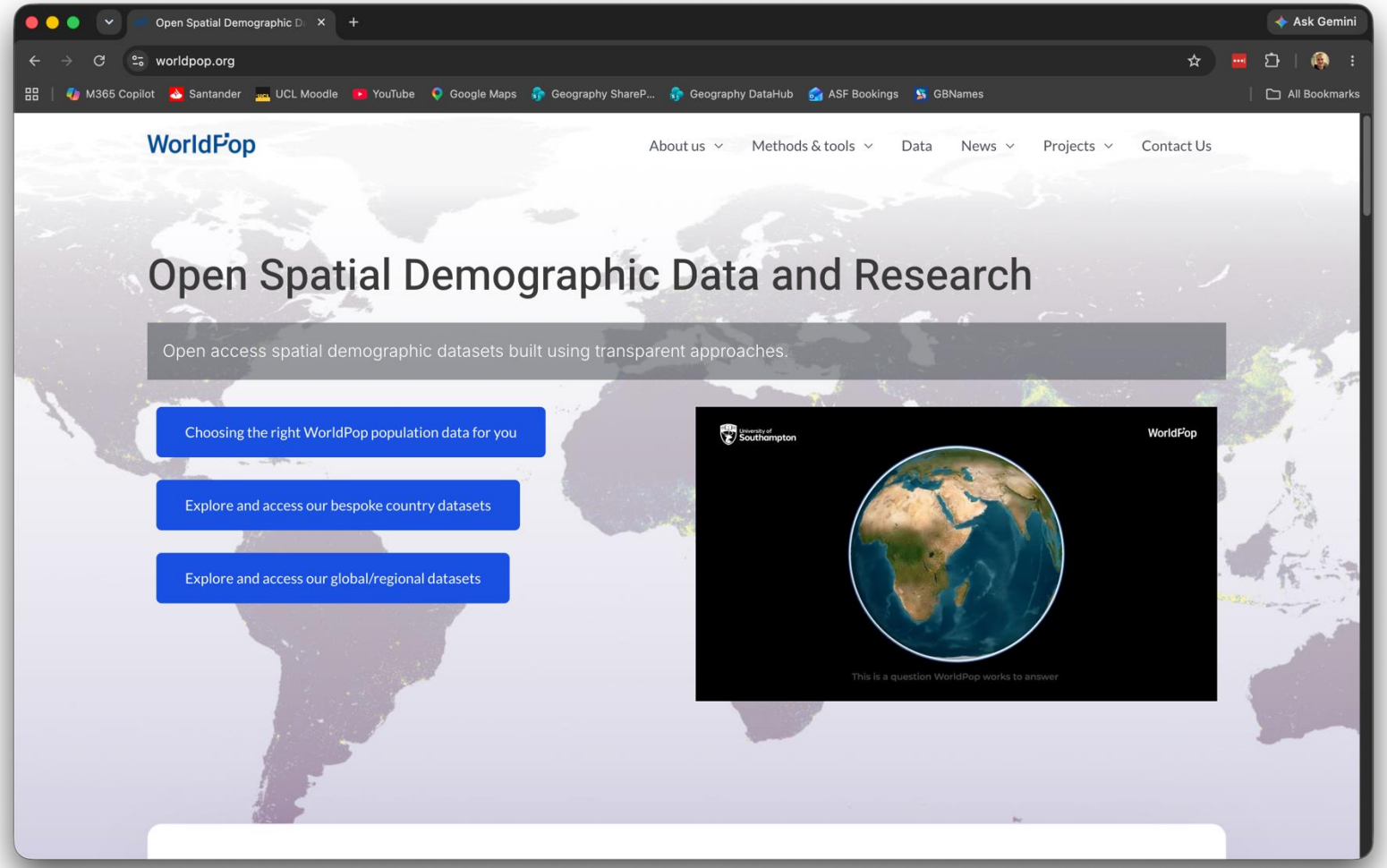
No clusters of one colour remain.

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

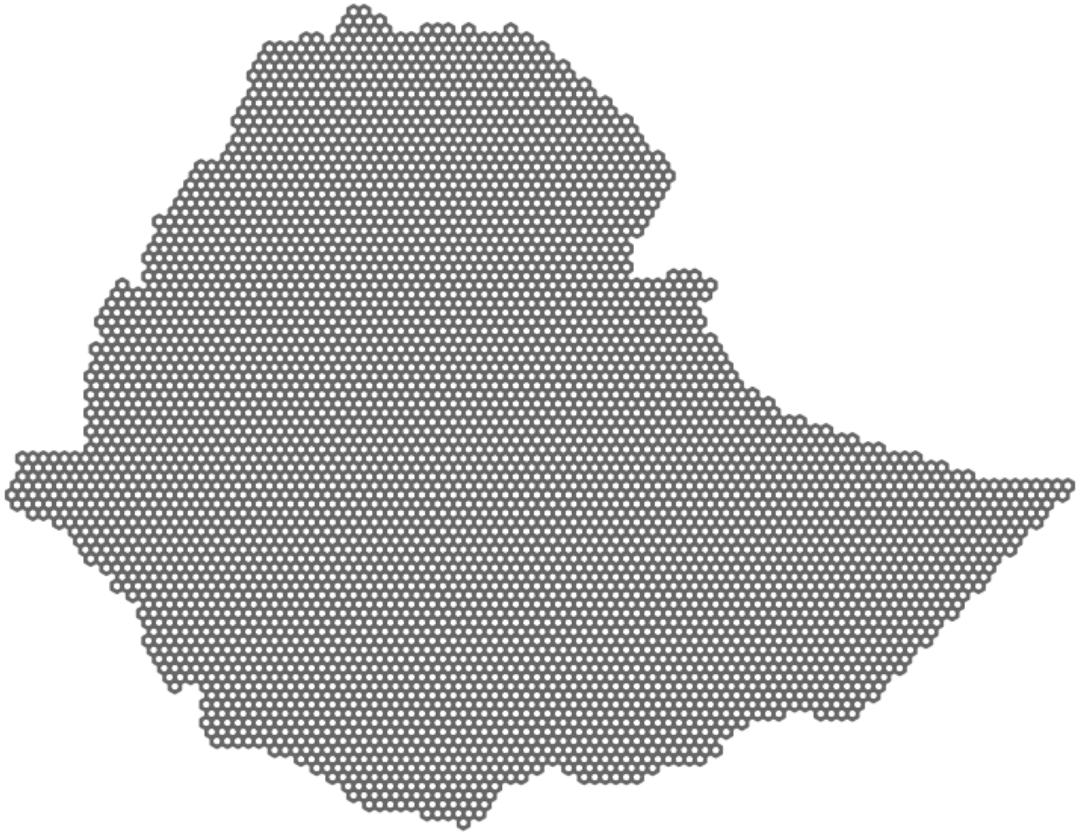
Raster data


Population count
0 to 20,000
20,000 to 100,000
100,000 to 250,000
250,000 to 500,000
500,000 to 1,000,000
1,000,000 to 2,500,000

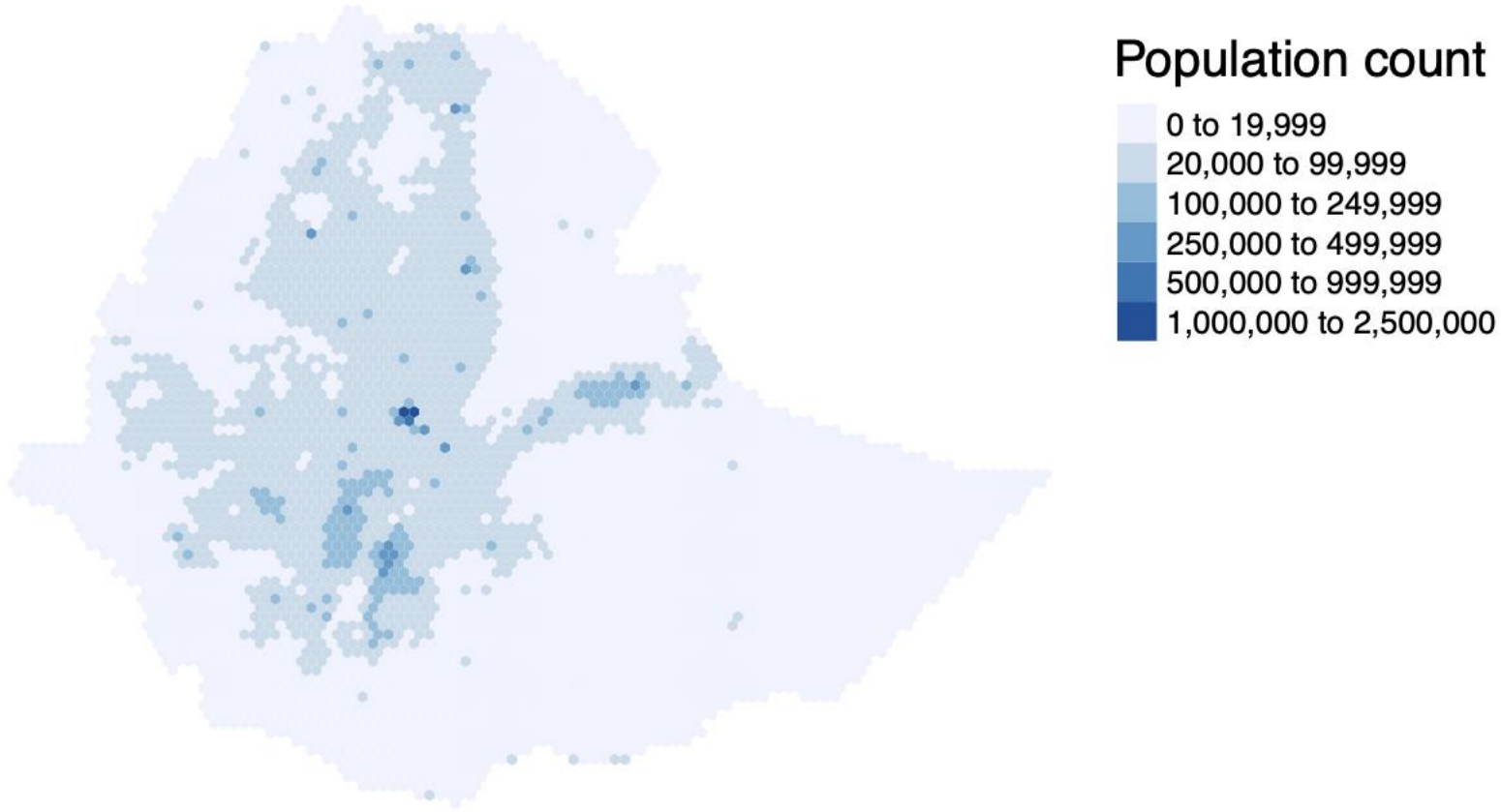
Raster data


Open Spatial Demographic D
Ask Gemini
worldpop.org
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
WorldPop
About us
Methods & tools
Data
News
Projects
Contact Us
Open Spatial Demographic Data and Research
Open access spatial demographic datasets built using transparent approaches.
Choosing the right WorldPop population data for you
Explore and access our bespoke country datasets
Explore and access our global/regional datasets
University of Southampton
WorldPop
This is a question WorldPop works to answer

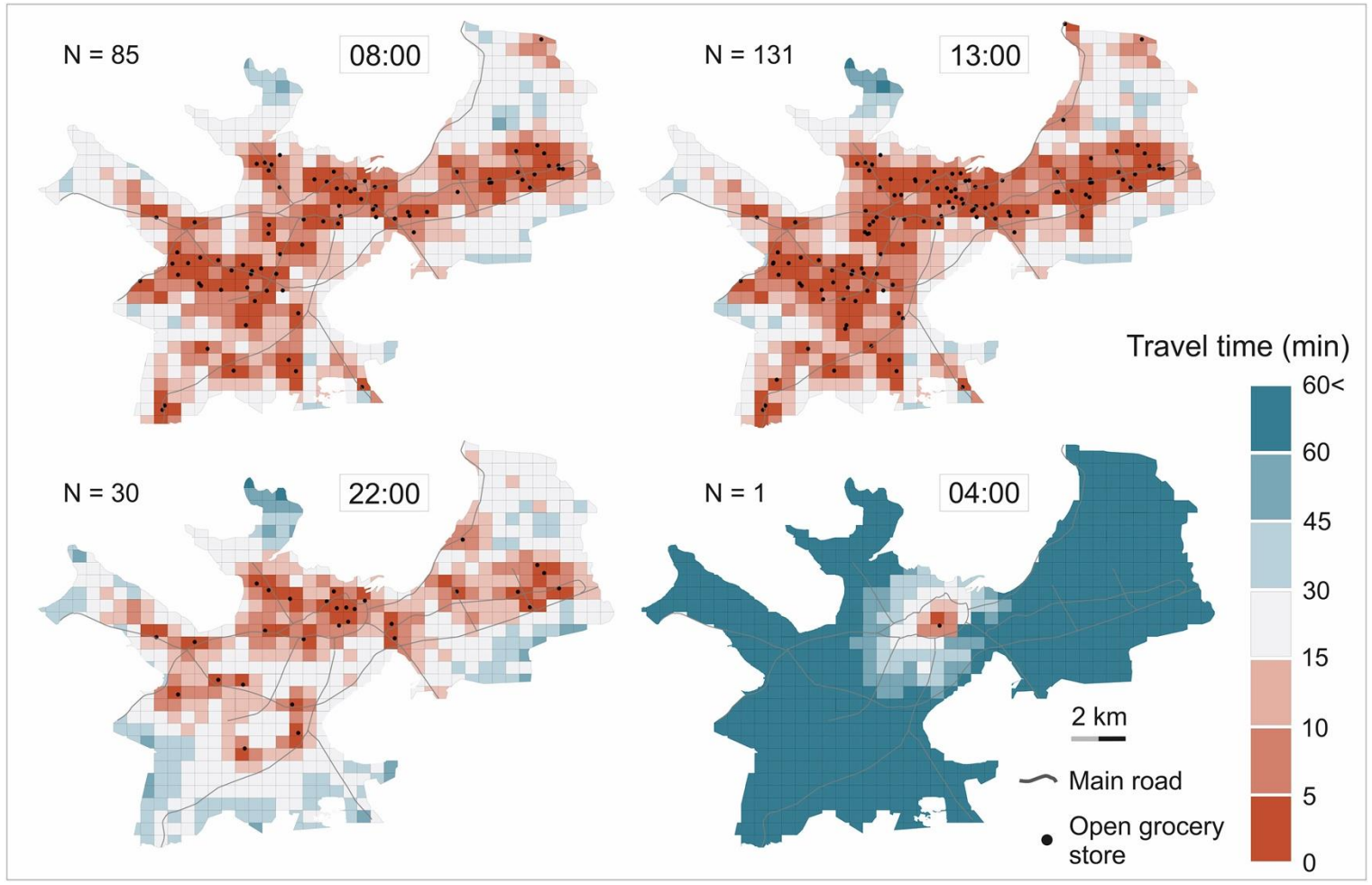
Raster data



Raster data

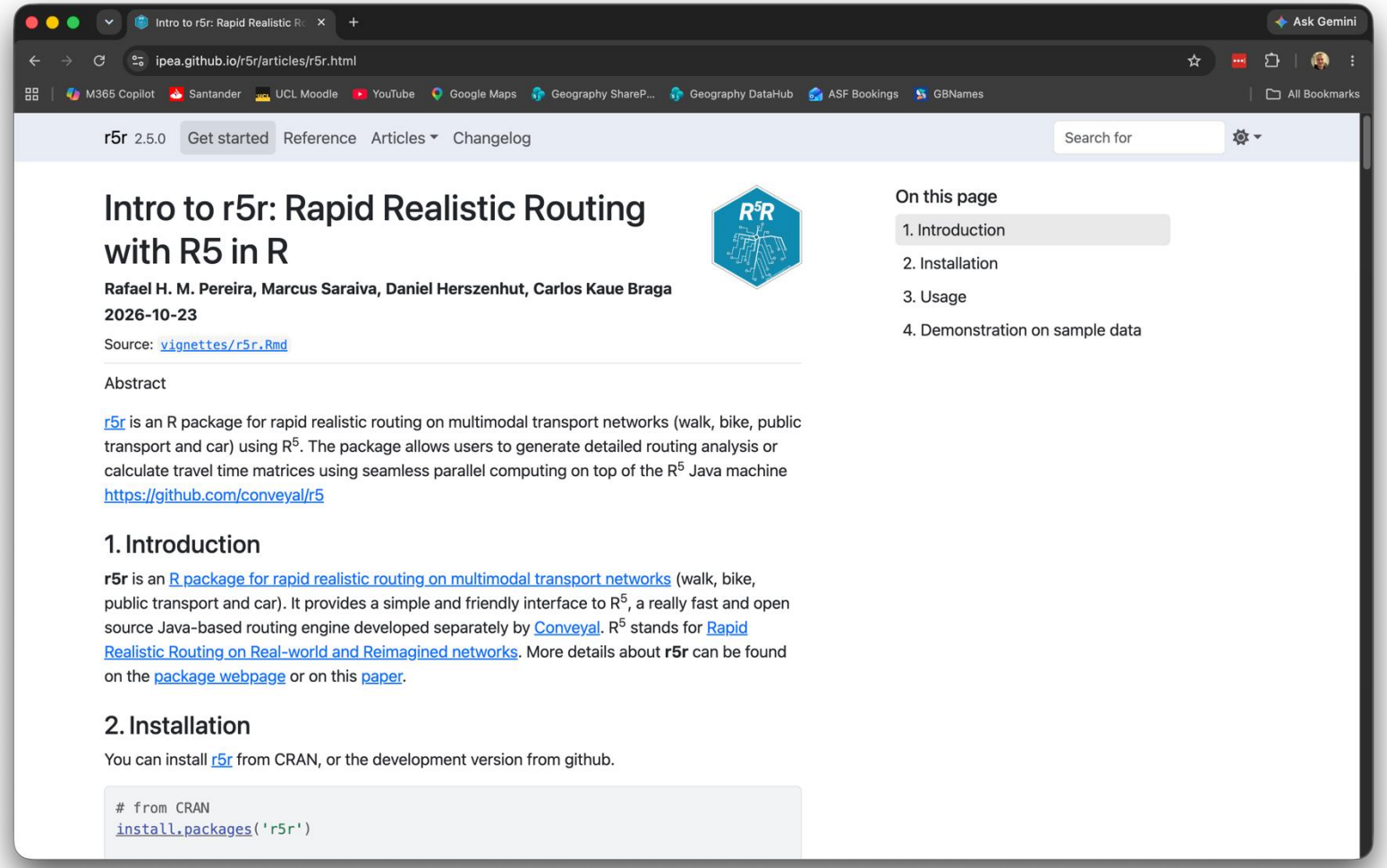

Population count
0 to 19,999
20,000 to 99,999
100,000 to 249,999
250,000 to 499,999
500,000 to 999,999
1,000,000 to 2,500,000

Accessibility analysis

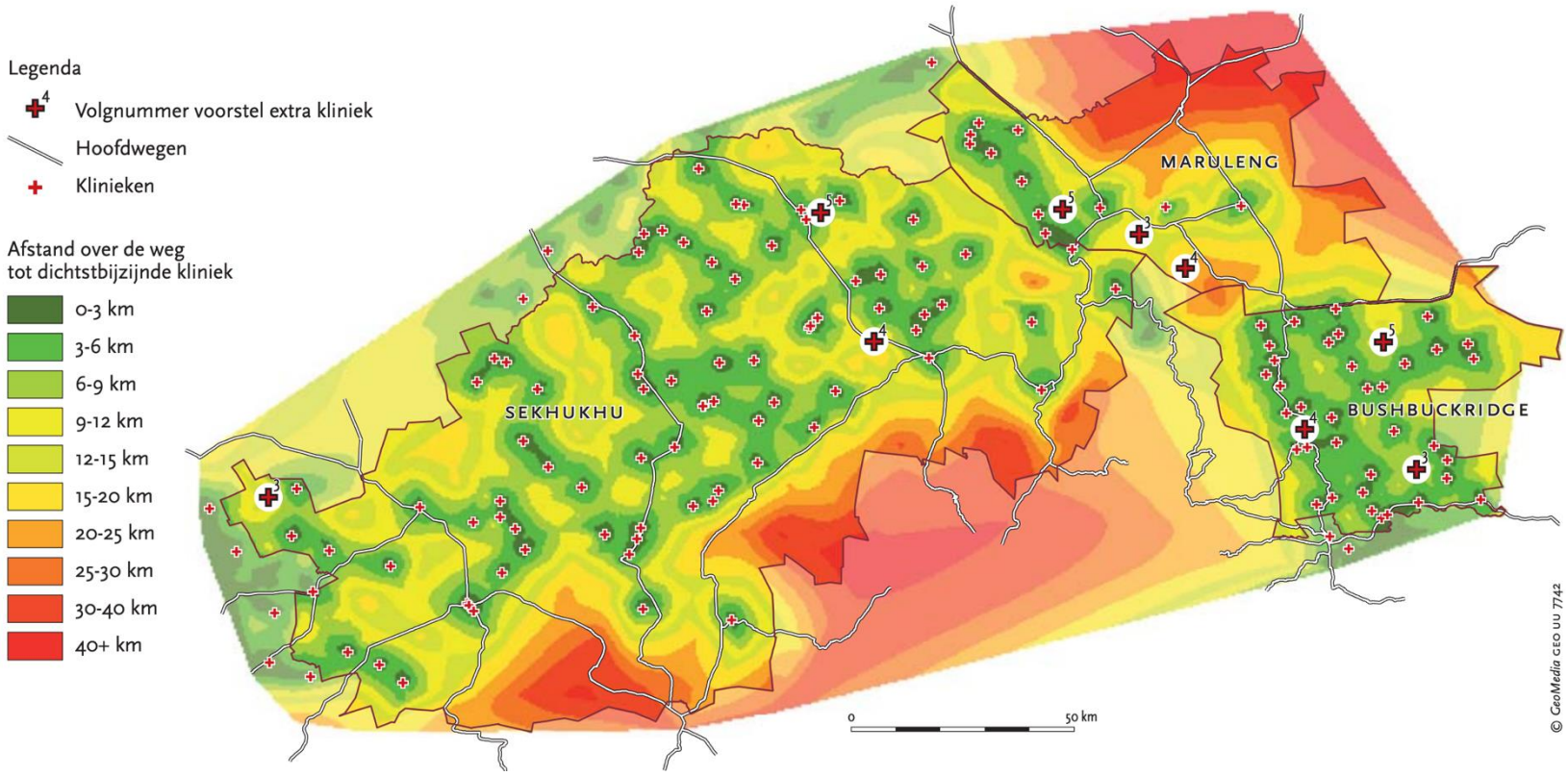


Järv *et al.* 2018

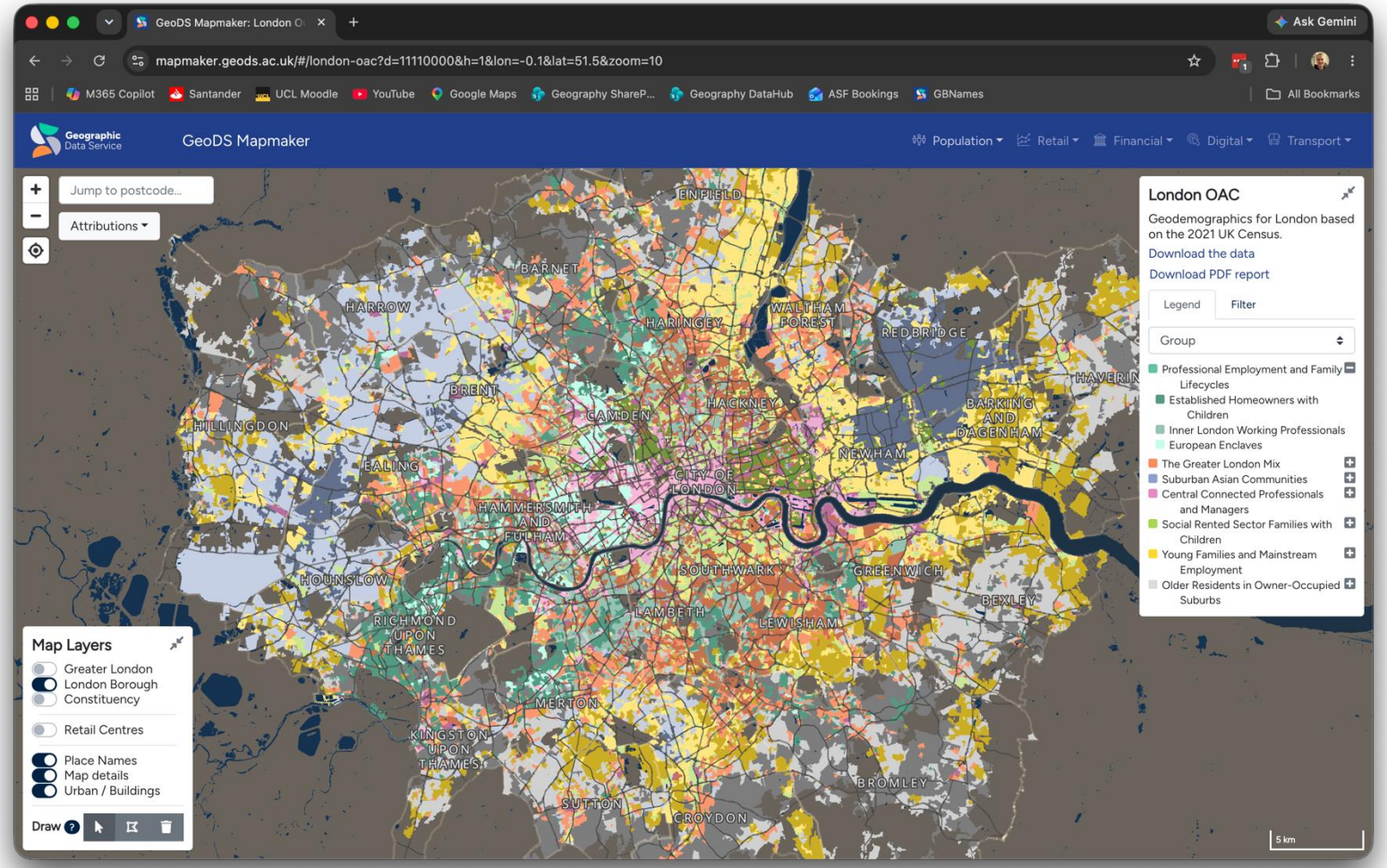
Accessibility analysis



Accessibility analysis



Geodemographic classification



2021 London Output Area Classification on mapmaker.geods.ac.uk

Novel data sources

Traditionally, quantitative datasets in geography and the social sciences:

- Are collected for a specific purpose, following a careful study and design.
- Contain very detailed information on a particular topic.
- Are of high quality and of known provenance.

Novel data sources

Traditionally, quantitative datasets in geography and the social sciences are:

- Expensive to produce (e.g., Census, longitudinal surveys).
- Of limited spatial granularity (privacy preserving).
- Infrequently updated.

Novel data sources

- New sources of data: the digital exhaust.
- Diverse in quality and resolution.
- Arguably: higher spatial granularity, higher temporal granularity.
- Many geographical questions and applications although geographers are often not the ones providing answers.

Novel data sources



Summary

Summary

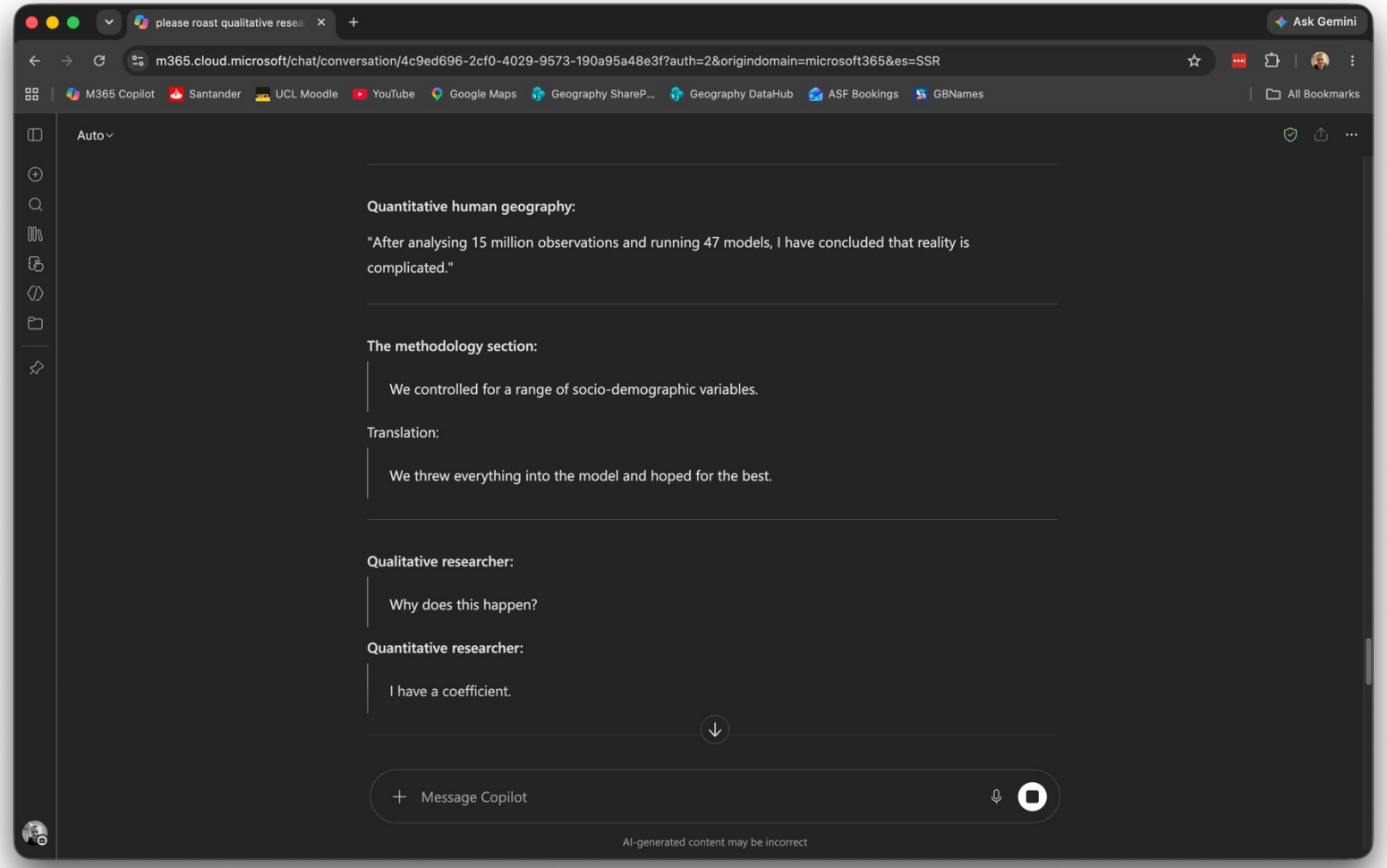
- Maps are important tools to effectively convey spatially heterogeneous information.
- At least two things are required: GIScience and GISystems.
- Essential to think about how to present your data.
- There is no one map to rule them all.
- Lots more quantitative spatial analysis out there, much of it in Geocomputation.

Module summary

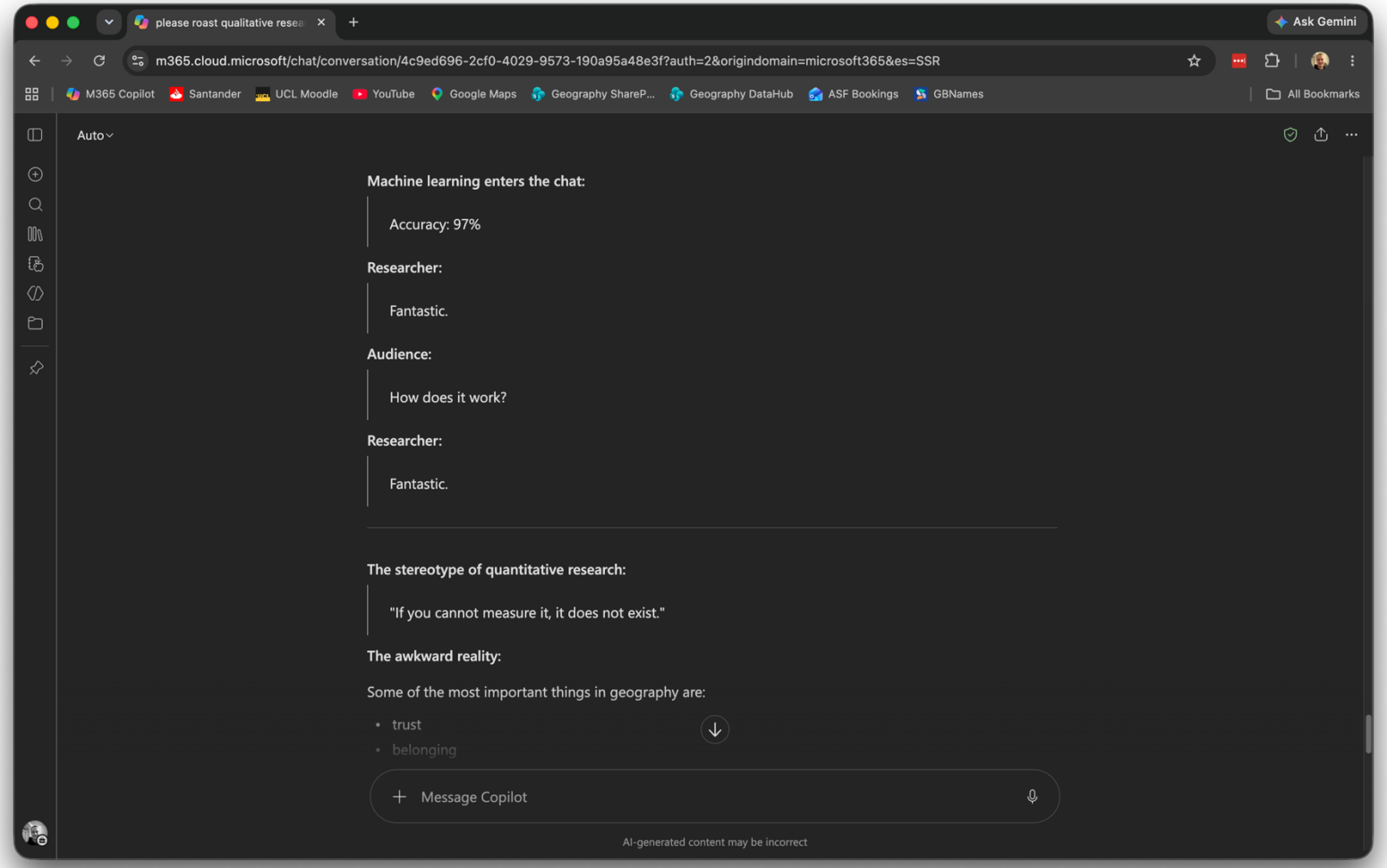
What we covered

- R and descriptive statistics.
- Sampling, confidence intervals and hypothesis tests.
- Chi-square, correlation and regression.
- Spatial data, attribute joins and first maps in QGIS.
- Mapping, aggregating points to areas and more spatial analysis.

Quantitative methods



Quantitative methods



Questions

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Happy winter break!